

Context selection as a partial order: A Blackwell framework and verified LLM evidence

Andriy Bilous¹, Vasyl Lytvyn¹,
Petro Pukach^{2,*}, Zoriana Rybchak¹

¹ Department of Information Systems and Networks,
Institute of Computer Science and Information Technologies,
Lviv Polytechnic National University, S. Bandery St. 12, 79013 Lviv, Ukraine;
andriy.bilous@uitware.com (A.B.); vasyl.v.lytvyn@lpnu.ua (V.L.);
zoriana.l.rybchak@lpnu.ua (Z.R.)

² Institute of Applied Mathematics and Fundamental Sciences,
Lviv Polytechnic National University, S. Bandery St. 12, 79013 Lviv, Ukraine

* Correspondence: petro.y.pukach@lpnu.ua

Abstract

Prompt context is usually chosen by scoring each source with one number and keeping the top few. We study the limits of task-independent scalar rankings for context sources. Two sources can be *incomparable* in Blackwell’s sense; we prove that when they are, no single per-source score ranks them correctly for every task. That concerns the ideal user of a source; what a fixed model does with one is separate, and we measure it. Running each source through a battery of pass/fail tasks and comparing pass rates with exact confidence intervals, on six models from four vendors we verify at $\geq 90\%$ in both directions that two hand-built sources cross in realized pass rate. On a designed trap a lexical score, two dense retrievers and a cross-encoder prefer the wrong source, while a listwise LLM reranker reading the deciding detail does not. A strict superset of a sufficient source also lowers pass rates, verifiably on all twelve cells and as far as 100% to 0%; a character-matched padding arm and a reversed-order arm leave that collapse unexplained. Results hold for the evaluated snapshots, prompts and decoders on a hand-built battery; scaling the implied rule to large corpora remains open.

Keywords: context selection; Blackwell order; comparison of experiments; incomparability; partial order; retrieval-augmented generation; large language models; distribution-free verification; anti-monotonicity; prompt engineering

1 Introduction

Suppose a coding assistant is handed one of two short notes about a software module. The first note says how to *call* its main function: the order of the arguments, a keyword that is mandatory, and the error it raises on failure. The second note says how the module *encodes* numbers: a digit format ending in a checksum. For a task that calls the function the first note is decisive and the second is useless; for a task that encodes a number it is the reverse; and neither note can be reconstructed from the other. The two notes are *incomparable* in the sense of Blackwell’s comparison of experiments [1], and the consequence is immediate: no single relevance score ranks them correctly for every task,

because a score must crown one winner and here neither wins outright. In the body these are the sources W_1 and W_2 .

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information, $\mathcal{V}$ -usable information [2], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. This paper asks what a task-independent scalar ranking can and cannot represent, and answers it with a classical apparatus from statistical decision theory.

What is and is not new. The mathematics is classical: the garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam’s deficiency. So are several of the phenomena, each in its own literature: that utility rather than topical relevance is the right target [3] and is model-dependent [4]; that adding sources can lower accuracy while coverage saturates [5]; that rerankers are steered by lexical similarity [6]; that keeping a non-dominated set is already an operator over heuristic scores [7]; and that the Blackwell and Le Cam apparatus was brought to inference-time context selection concurrently with this work, over which we claim no priority [8]. Section 2 treats each in turn. The gap is that none of them gives the order itself an object or a test: cross-task non-transfer is described as a nuisance rather than read as a partial order, and no finite-sample procedure decides on a live model whether two particular sources are incomparable. That is what C1–C3 below supply.

One gap needs care. Blackwell’s theorem is about the *best possible* use of a source, but every number we measure comes from one fixed model: $\text{PASS}_M(W, T) = \mathbb{E}[v_T(\delta_M(T, w))]$, the pass rate of a single rule δ_M that is not optimal. So we state the order twice, for the ideal user and for the fixed model, and prove a one-way bridging lemma whose hypotheses are assumed, not measured; our sources’ structural incomparability rests instead on their construction (see Section 4); and we call our estimator a *value-deficiency surrogate*, not a Le Cam deficiency (see Section 6).

Contributions. The paper makes one claim, that the order over context sources is *partial* rather than total, and contributes the three pieces that claim needs: what follows from it, the instrument that makes it testable, and the evidence that it holds.

- (C1) **Theory: the order is partial, and no single per-source score survives it** (see Sections 4 and 5). We formalize context sources as statistical experiments on the task’s latent requirement, instantiate the Blackwell garbling order with its for-all-tasks dominance guarantee via the Blackwell–Sherman–Stein theorem, define the δ_M -restricted order the experiments probe, and prove a one-way bridging lemma between them. What follows from taking *incomparability* as the object is an impossibility: Blackwell-incomparable sources cannot be ranked by *any* single per-source score $\varphi(W)$, relevance or utility, for all tasks (Theorem 2), with the query-conditioned topical case isolated separately (Proposition 2). Information retrieval has long argued that no single fixed document score is optimal once documents interact (see Section 2.1); we derive that as a theorem in the Blackwell frame and name the exact structural condition under which it bites.
- (C2) **Verification: a measurable deficiency surrogate with finite-sample guarantees** (see Section 6). We define the *population* gap $d_{M, \mathcal{D}}(W_1, W_2) = \max(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)])$ together with its plug-in estimate $\hat{d}_{\mathcal{D}}$, a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform

Table 1: Glossary: the recurring formal terms and their engineering reading.

Formal term	Engineering reading
source W (an experiment)	a candidate context item: a doc, note, or module you could paste into the prompt
Blackwell dominance $W \succeq_B W'$	W answers everything W' answers; keeping W and dropping W' is safe
incomparability	each source is decisive on tasks where the other fails; no single number can rank them once and for all
garbling	W' is a lossy transformation of W : everything in W' is derivable from W
anti-monotonicity ($\Delta_T < 0$)	pasting more true, on-topic text into a working prompt can break it
non-dominated front	the shortlist you must keep whole, because nothing on it verifiably covers anything else on it
verification	a claim measured with exact finite-sample error bars, still significant after multiple-comparison correction, under the per-cell Bernoulli sampling model of Assumption 2
model rule δ_M	the model as it actually behaves on the prompt, not an idealized optimal user of the information
gap $d_{M,\mathcal{D}}$, plug-in $\hat{\delta}_{\mathcal{D}}$	how many PASS points you lose by using W in place of W' on the task family; the hat marks the sample estimate, never the population gap

verification: exact Clopper–Pearson intervals [9] at Bonferroni level [10] $\alpha = \eta/(2|\mathcal{D}|)$, valid jointly on a single event, so that $L_{\mathcal{D}} > 0$ verifies a positive value gap at confidence $\geq 1 - \eta$. The selection multiplicity is paid explicitly rather than absorbed into a single-sample sup inequality [11], which does not cover an argmax over distinct cells. This is what makes the partial order testable on a real model behind a real verifier.

(C3) Evidence: verified incomparability and scalar-ranking failure on live models (see Section 7). Verified empirical incomparability on six models across four vendors; the ranker-spectrum mis-rank on a confound-controlled trap (lexical cosine, dense bi-encoders, and a cross-encoder all fail; an LLM listwise reranker that reads the decisive content escapes, exactly the topicality boundary the theory draws); and verified anti-monotonicity, a strict superset of a sufficient source collapsing PASS, with order, length-padding, and routing-instruction controls, all with the scope limits stated in Section 8.4.

How to read the formal terms, and where things are. Table 1 gives the engineering reading of each recurring term, and a reader who wants the measured result first can go straight to Section 7. Otherwise the order is prior work (see Section 2), the partial order and the two impossibility results (see Sections 4 and 5), the verification (see Section 6), the six-model measurement (see Section 7), scope and conclusions (see Sections 8 and 9), with proofs, the per-cell record, the diagnostics and the protocol in Appendices A–D.

2 Related Work

2.1 Scalar selection and its limits in information retrieval

Context and demonstration selection is dominated by *scalar* criteria, each inducing a total order and a top- k prefix: topical relevance, lexical or learned [12]; $\mathcal{V}$ -usable information [2, 13]; expected information gain and Bayesian experimental design [14, 15]; Shapley- and influence-based valuation of demonstrations [16, 17] and of retrieved documents [18]; perplexity-reduction utility [19]; and binary sufficient-context detection [20]. Each collapses the comparison of two sources to one real number, so each is an instance of the excluded class rather than a competitor: Theorem 2 covers the query-independent ones, and Proposition 2 the topical query-conditioned ones that meet its hypotheses. For the closest neighbour, $\mathcal{V}$ -usable information, we prove Blackwell-monotonicity (Proposition 1); for mutual information and fixed-prior EIG it follows from data processing; for a fixed decoder’s perplexity-gap we claim nothing (Remark 5). Calibration does not change the class: conformal context filtering [21] calibrates a threshold *on* a scalar and inherits its total order, and scorers of equal ranking quality can retain different document sets [22]. Identifying a non-dominated set from noisy samples is classical in the bandit literature [23]; what is new here is the object it runs on, (model, verifier, task-battery) triples, not the statistical machinery.

Retrieval has itself moved from relevance to *utility* [3], which is model-specific [4] and depends on which other passages are present [24]. That is the same diagnosis from the other side, and this paper is its formal boundary. Utility re-measured per task by running the model is not a fixed per-source score but the probe our estimator is built from; utility *predicted* by a learned scorer is a number again, and falls under whichever of the two results applies. Reading the decisive coordinate is one way out, the listwise boundary; reading public task metadata is another (Remark 6). Three families sit there. Listwise rerankers [25] and learned set selectors [26, 27] are query-conditioned *set* functions, outside the theorem’s scope but carrying no dominance guarantee. Skyline selection [7] keeps the non-dominated set, the operator the partial order calls for, but takes dominance over heuristic per-item scores rather than task value. Diversity-aware selection, from MMR [28] to DPPs [29] and submodular selection [30], optimizes similarity geometry rather than verified value. Directed-information γ -covering [31] is closest in form, its covering relation being asymmetric and information-theoretic; but it orders redundancy *between sources*, computed query-agnostically, so it prunes what another chunk predicts and never exhibits two sources each decisive where the other fails.

The limits of scalar relevance are among the oldest themes in the field, and we sit inside that lineage: the Probability Ranking Principle [32] is optimal only under independence, its failure once documents interact is long documented [33], and interdependence was given ranking principles of its own in portfolio theory [34] and the quantum PRP, whose “interference” term names exactly this dependence [35]. The LLM-era version, relevance-maximizing retrieval degrading answers, is measured directly [36], and the observation that relevance does not imply *sufficiency* is argued at the set level [37]. Our delta is the form of the statement: a theorem in the comparison-of-experiments frame, with the structural condition under which every query-independent scalar must mis-rank, and a verified instance on six production models.

2.2 Task-dependence and context degradation

That added context can degrade performance is by now well documented: a single irrelevant sentence tanks reasoning [38]; position matters [39]; high-scoring distractors hurt while random noise can help [40]; retrieval curves are inverted-U in passage count [41]; document count hurts at fixed length [42]; added length alone hurts [43]; and the effect persists across model generations [44]

and agent-scaffold components [45]. The genus is older than language models: in data integration, adding sources raises coverage while integration quality first rises and then falls, so that “less is more”: low-quality sources deteriorate the fused result rather than bringing the expected gain, and the right move is to select a subset before integrating [5]; the lesson has since been carried explicitly to augmented language models [46]. We do not claim to discover this genus. Two differences locate our finding inside it. Where data integration traces the harm to low-quality sources and restores monotonicity by redesigning the fusion rule, our two sources are individually correct and sufficient, the harm survives gross-length, block-order and formatting controls on the one model where those controls were run and is consistent with convention routing, and the rule δ_M is a fixed model that cannot be redesigned, so the response is selection rather than fusion. And where “interference” in the mechanistic RAG literature means distractor passages steering a model, a failure recently localized and corrected in a sparse feature space [47], and inter-context *conflict* means contexts that contradict each other [48], our sources neither distract nor conflict: both are true and both are needed, and the superset still collapses. Our anti-monotonicity finding contributes the order-theoretic form of this: a strict *superset* of a sufficient source, on sources separately *verified* Blackwell-incomparable, collapses PASS under a per-task verification of the direct contrast $\text{PASS}(W_{1+}, T) < \text{PASS}(W_1, T)$: a reversal no monotone context-value scalar can carry. A representation-level account of when context helps, an additive error decomposition with alignment conditions [49], is complementary: even in its multi-context extension it characterizes *when* a context corrects the error, not an order relation between candidate sources. Pareto-incomparability of prompts along competing metrics *within* a single task [50] is orthogonal to our object, incomparable *sources across tasks*.

2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has entered machine learning and, concurrently with this work, inference-time context selection itself. Alpila [8] formalizes the admission problem against a *frozen decoder* as minimization of Le Cam deficiency, pairs it with the Vereshchagin–Vítányi conditional structure function for the representation problem, proves a capacity floor below which the deficiency cannot be driven, and validates a curve-aware allocator on seven decoders in four families; a plain-language companion states the framing [51] and the artifact is released [52]. Our object is different, the order *between two admitted sources across tasks*: we show it is partial by *measuring* mutual non-domination, derive that no single per-source score can be faithful to it, and verify a strict superset of a sufficient source collapsing PASS, with gross-length, block-order and routing controls run on one model, not every confound excluded. Incomparability in this sense cannot arise inside a single declared decision problem, so its absence there is a different question rather than a gap.

A second line is training-*dataset* valuation. Zheng et al. [53] order datasets by Blackwell informativeness and show that the mutual information between the curated and the test data is a *strategy-proof* score: a curation step that garbles the data, so that $\theta \rightarrow D \rightarrow f(D)$ is a Markov chain, cannot raise it. That is the data-processing direction, and it is one-directional; a scalar can be monotone along the order without representing it, so their result is compatible with ours rather than opposed to it. Their comparison is also taken against the Bayes-optimal user of the data, ours against a fixed model rule on a task family. What Theorem 2 adds is the converse question, whether any query-independent scalar can be *faithful* to the order, which a monotone score need not be once the order is partial. Nearby uses stop short of what we do: garbling to harden LLM evaluations [54], statistical deficiency to order training *tasks* [55], a deficiency-induced *directional* dominance in transfer learning (no incomparability, no distribution-free bound, outside language modeling) [56], comparison-of-experiments ideas in loss design [57], and a survey of Blackwell’s

theorems for AI [58]. Another inference-time application sits on the other side of the model: Zhang et al. [59] rank the information structures induced by multi-agent protocols, showing that voting and debate are no more informative than the agents’ pooled private information and identifying Bayesian pooled-posterior maximisation as the Blackwell upper bound on any such protocol. That is a dominance statement about how several models’ signals combine; ours is an incomparability statement about which single source one model is given, so the two orders are taken over different objects and neither implies the other.

2.4 Where the two results land in the design space

Sorting the methods above on three axes, whether a method is cheap enough to run at retrieval scale, whether it conditions on the task, and whether it can represent a non-dominated *set* rather than a ranking, makes the pattern uniform. Every method cheap enough to precompute emits one number per item, however that number is produced: late interaction over multi-vector representations [60] and retrievers supervised by the language model itself [61, 62] are scored per item no less than lexical relevance is. All of them therefore fall under Theorem 2 when that number is fixed across tasks. The methods that escape do so either by reading the decisive content per candidate, which costs a model call, or by routing on public task metadata, which Proposition 2 does not exclude (Remark 6). No prior family is both cheap and faithful to the partial order; that empty cell is the open problem this paper isolates; the verified non-dominated set pays probe cost to sit in it, and what that cost buys is a verified relation *between* the candidates, not a guarantee about any selector built on the set.

3 Formalization

We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task’s latent requirement. This section fixes the objects and separates two decision rules that the informal literature silently conflates: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which every empirical quantity in this paper measures.

3.1 Tasks, latent requirements, and context sources

A *task* T fixes a latent requirement S_T , the object a deterministic verifier v_T checks (e.g. the intended program behaviour). We take S_T to be a *vector of coordinates*

$$S = (S_1, S_2, \dots, S_m) \in \mathcal{S} = \mathcal{S}_1 \times \dots \times \mathcal{S}_m, \quad (1)$$

where each coordinate S_j encodes one logically independent requirement. Modelling S as a product of coordinates rather than an opaque atom is what makes the experiment model well posed for fixed-string sources (Section 3.2, Lemma 1).

Definition 1 (Context source as an experiment). A *context source* W is a Markov kernel $\kappa_W : \mathcal{S} \rightarrow \Delta(\mathcal{W})$, i.e. a statistical experiment indexed by the latent parameter S , emitting a signal $w \sim \kappa_W(\cdot \mid S)$ in a signal alphabet $\mathcal{W}$ (the source’s text). The experiment is identified with its kernel; the sample space is $\mathcal{W}$ and the observed datum is the supplied text w .

Definition 2 (Decision rules and PASS). A *decision rule* is a (possibly randomized) map $\delta : (T, w) \mapsto \hat{S}$ producing a candidate solution. Utility is the binary verifier $u(\hat{S}, S) = v_T(\hat{S}, S) \in \{0, 1\}$,

which scores the candidate against the convention in force: the same answer passes under one convention and fails under another, which is why the second argument cannot be dropped. Two decision rules matter:

- (i) the *model rule* δ_M realized by a fixed language model M at a fixed decoding configuration θ (model weights and sampling parameters; see Section 3.3). Its realized pass rate on task T under source W is

$$\text{PASS}_M(W, T; s) := \mathbb{E}[v_T(\delta_M(T, w), s)], \quad w \sim \kappa_W(\cdot | s), \quad (2)$$

written $\text{PASS}_M(W, T) := \text{PASS}_M(W, T; S_T)$ at the single realized convention S_T , which is the quantity every measurement in this paper reports;

- (ii) the *Bayes rule* δ_T^* , the utility-optimal rule for task T under a prior μ_T on $\mathcal{S}$ (the task family’s distribution over private conventions), achieving the optimal value

$$V^*(W, T) := \sup_{\delta} \mathbb{E}_{S \sim \mu_T} \mathbb{E}[v_T(\delta(T, w), S)], \quad w \sim \kappa_W(\cdot | S). \quad (3)$$

Remark 1 (The two quantities are taken under different measures). PASS_M in (2) is *conditional on one latent state*: the expectation runs over the source kernel and the decoder with S held at the realized convention S_T . V^* in (3) is a *Bayes* value, averaged over μ_T . Consequently $V^*(W, T) \geq \text{PASS}_M(W, T)$ is *not* automatic from optimality, and we never use it in that form. The obstruction is elementary: let S be a fair bit, let W be uninformative, and let v_T reward naming S ; the constant rule “answer 0” has $\text{PASS}_M = 1$ at the state $S = 0$ while $V^* = 1/2$. Nor can the gap be closed by taking μ_T to be a point mass at S_T : every guessing cap $V^*(W, T)$ would then equal 1 and the caps of Lemma 2 would lose all content. What optimality does give is the averaged statement $V^*(W, T) \geq \text{PASS}_M(W, T) := \mathbb{E}_{S \sim \mu_T}[\text{PASS}_M(W, T; S)]$. Lemma 2 is therefore proved through the sufficiency of the winning source rather than through any such comparison; the measured pass rates enter its proof nowhere, and neither bound a Bayes value nor estimate a guessing cap.

The distinction in Definition 2 is load-bearing. Blackwell’s theorem (see Section 4) is a statement about V^* ; every number we report, PASS_M , the deficiency estimator of Section 6, the interference statistic of Section 6.3, and the entire six-model program of Section 7, is a statement about the single fixed rule δ_M . We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*. Reading $\kappa_W(\cdot | S)$ as a constant kernel δ_{w_0} independent of S is degenerate: such a kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial σ -algebra, which would make any garbling argument vacuous. We instead model each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

Definition 3 (Projection sources). A source W that reveals the coordinates indexed by a set $A \subseteq \{1, \dots, m\}$ is the deterministic kernel

$$\kappa_W(\cdot | S) = \delta_{\pi_A(S)}, \quad \pi_A(S) := (S_j)_{j \in A}, \quad (4)$$

i.e. supplying W lets a decision-maker condition on $\pi_A(S)$ and on nothing outside A . Write $\mathcal{F}_W := \sigma(\pi_A)$ for the sub- σ -algebra of $\mathcal{S}$ generated by the projection. In our instantiation W_1 reveals coordinate S_1 (the call contract, $A_1 = \{1\}$), W_2 reveals S_2 (the wire invariant, $A_2 = \{2\}$), and the superset $W_{1+} = W_2 ++ W_1$ reveals both ($A_{1+} = \{1, 2\}$).

Under Definition 3 the informativeness of a fixed string is exactly the resolving power of its σ -algebra $\mathcal{F}_W$. The signal alphabet is the range of the projection, and the text “varies with the task” because the realized coordinate values $\pi_A(S_T)$ differ across task instances T . This is what makes the comparison of Section 4 a comparison of experiments rather than a metaphor, and it is what a rigorous treatment of deterministic sources requires [63].

3.3 The conditioning parameter θ

The model rule δ_M carries an implicit parameter θ : the model’s weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write $\text{PASS}_M(\text{none}, T)$, the model’s no-context pass rate, for the part of a private convention that θ already encodes; the value of a source is model-relative through it. When θ already “knows” a convention (the model can guess it without the source), that baseline is high and W ’s marginal contribution collapses; when the convention is genuinely unknown to θ , the baseline floors and the source has to carry the cell. We deliberately do *not* write this as a conditional mutual information $I(S; W \mid \theta)$: under Definitions 1 and 2 the law of (S, W) is μ_T followed by κ_W and carries no θ , so conditioning on a fixed model index would leave the mutual information unchanged and model-*independent*, the opposite of what is measured here; a genuinely model-relative information quantity would need a subjective prior we neither define nor measure. This θ -dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in Section 7, where the guessability of a fixed convention spans 0% to 100% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by θ ; we suppress it except where the model-relativity is the point.

4 The Blackwell order over context sources

With sources modeled as experiments (see Section 3), the classical comparison of experiments applies to them directly: we state the order, record that it is partial, and bridge its two registers.

4.1 The order and its two registers

Definition 4 (Garbling order). $W_1 \succeq_B W_2$ (W_1 *Blackwell-dominates* W_2) iff there is a Markov kernel (garbling) $G : \mathcal{W}_1 \rightarrow \Delta(\mathcal{W}_2)$ with $\kappa_{W_2} = G \circ \kappa_{W_1}$; i.e. W_2 ’s signal is a noised, post-processed version of W_1 ’s, computable from W_1 ’s signal alone without further access to S .

Theorem 1 (Blackwell–Sherman–Stein; universal dominance for the optimal rule). *Let W_1, W_2 be experiments on a common finite parameter space $\mathcal{S}$ with standard Borel signal spaces. The following are equivalent (the implication (i) $\Rightarrow$ (ii) needs neither finiteness nor further regularity, and is the only direction Corollary 1 and Lemma 2 use):*

- (i) $W_1 \succeq_B W_2$ (Definition 4);
- (ii) *for every prior μ on $\mathcal{S}$, every finite decision space, and every bounded loss ℓ , the Bayes risk under W_1 is no larger than under W_2 ; equivalently the optimal value satisfies $V^*(W_1, T) \geq V^*(W_2, T)$ for every task T with bounded utility.*

Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii).

The theorem is classical [1, 63] and we use it as a black box. (i) $\Rightarrow$ (ii) holds for arbitrary $\mathcal{S}$ with no further regularity, by simulating the garbling. We invoke the converse (ii) $\Rightarrow$ (i) once only, in

Theorem 2, and there for a finite state space, the case Blackwell [1] proves and our coordinate construction realizes; on a general state space the converse needs the convex closure taken in the definition of more-informativeness [64]. Reading “experiment” as a context source (Definition 1) is faithful because the kernels of Definition 3 are standard Borel, which says nothing about the “decision problem” reading: clause (ii) quantifies over *all* bounded utilities, whereas the tasks we can execute are only the binary-verifier subclass $v_T \in \{0, 1\}$ of Definition 2. That restriction is harmless in (i) $\Rightarrow$ (ii) but not in the converse, whose witness need not be binary; it costs nothing for the projection sources of Definition 3, whose binary witnesses are exhibited in the proof of Lemma 2(a), and beyond them it is a scope limit of our presentation, not a property of the theorem.

Corollary 1 (Universal context dominance, C1). *If $W_1 \succeq_B W_2$ then, for the optimal user, supplying W_1 is at least as good as supplying W_2 on every task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information, $\mathcal{V}$ -usable information) provides.*

Remark 2 (Preorder on kernels; “partial” means not total). On the class of all experiments $\succeq_B$ is reflexive and transitive but not antisymmetric, since two experiments can garble each other without being equal, so it is a *preorder* and the induced partial order lives on Blackwell-equivalence classes [1]. Throughout, “partial” means *not total*: there exist experiments V, V' with neither $V \succeq_B V'$ nor $V' \succeq_B V$, and on that class the order is not even a lattice [63, 65]. (The bound variables here are not the paper’s named sources: our own projection family is better behaved, and the next subsection shows $\succeq_B$ is antisymmetric on it with the induced order the Boolean lattice on coordinate sets, Lemma 1(a).) Incomparability survives on that family and is exactly the regime our experiments target. One caution for what follows: PASS_M is *not* an invariant of a Blackwell-equivalence class, and it need not be: a rule keyed to surface form can prefer differently between two experiments that garble each other. The padding arm W_{pad} of Appendix B reveals no coordinate any verifier checks, so it is Blackwell-equivalent to W_1 , yet the two do not score alike: 40/40 against 31/40 on `api_argorder`, a difference whose two-sided 90% interval $[+0.0202, +0.3846]$ excludes zero, and 40/40 against 39/40 on `api_post_ok`, where it does not. One cell therefore certifies an instance and the other only illustrates the possibility. Either way the δ_M -restricted order of Section 4.3 is not read on the quotient.

4.2 Deterministic sources: incomparability via σ -algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated σ -algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

Lemma 1 (Refinement characterization for projection sources). *Let W_A, W_B be projection sources revealing coordinate sets A, B (Definition 3), with coordinates $S_1, \dots, S_m$ mutually independent under some full-support prior and each S_j non-degenerate. Then:*

- (a) $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$;
- (b) *consequently W_A and W_B are Blackwell-incomparable iff $A \not\subseteq B$ and $B \not\subseteq A$, i.e. each reveals at least one coordinate the other hides.*

Lemma 1 is the formal version of the garbling argument used in Section 7: the call contract (W_1 , revealing the private arg-order / required-keyword / exception-type facts S_1) and the wire format (W_2 , revealing the cents-scale / check-digit facts S_2) give $A_1 = \{1\}$, $A_2 = \{2\}$, and neither set contains the other, so W_1, W_2 are Blackwell-incomparable by construction (and W_{1+} , revealing $\{1, 2\}$, dominates both).

4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns V^* ; our experiments measure PASS_M for a single fixed, generally sub-optimal rule δ_M . An inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a δ_M -restricted order.

Definition 5 (δ_M -restricted, $\mathcal{D}$ -restricted order). Fix the model rule δ_M and a finite task set $\mathcal{D}$. Say $W_1 \succeq_{M,\mathcal{D}} W_2$ iff $\text{PASS}_M(W_1, T) \geq \text{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$. Call W_1, W_2 ($M, \mathcal{D}$)-*incomparable* iff $W_1 \not\succeq_{M,\mathcal{D}} W_2$ and $W_2 \not\succeq_{M,\mathcal{D}} W_1$, i.e. each strictly out-performs the other on some task in $\mathcal{D}$.

$\succeq_{M,\mathcal{D}}$ is the order our verification measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite $\mathcal{D}$); within those restrictions it is still a genuine, *for-all-tasks-in- $\mathcal{D}$* dominance statement, not a scalar.

Lemma 2 (One-way bridge under coordinate separation). *Neither order implies the other unconditionally; each direction of the bridge carries an explicit hypothesis.*

- (a) *Let W_1, W_2 be projection sources revealing coordinate sets A_1, A_2 (Definition 3), and let $T_a, T_b \in \mathcal{D}$ be tasks whose verifier-decisive coordinates lie in $A_1 \setminus A_2$ and $A_2 \setminus A_1$ respectively, each verifier being determined by its decisive coordinate together with facts known μ_T -almost surely, and μ_T -almost surely accepting some action measurable in that coordinate and those facts (sufficiency and solvability under the informed source; solvability is not implied by the dependence). Let the guessing caps $g_a := V^*(W_2, T_a)$ and $g_b := V^*(W_1, T_b)$ be the Bayes values (3) attainable without the decisive coordinate. If $g_a < 1$ and $g_b < 1$, so that neither deprived source determines the coordinate its task turns on, then W_1, W_2 are Blackwell-incomparable. The caps are hypotheses of this criterion, not quantities the experiments determine. Per cell, Section 7 reports a crossover between the same fixed rule's realized rates, $\text{PASS}_M(W_1, T_a) > \text{PASS}_M(W_2, T_a)$ and $\text{PASS}_M(W_2, T_b) > \text{PASS}_M(W_1, T_b)$; since PASS_M is state-conditional and V^* a Bayes value, such a comparison neither estimates nor bounds g_a or g_b , though it is consistent with $g_a, g_b < 1$ (Remark 1); we nowhere report the cap conditions as empirically verified.*
- (b) *The converse fails in general: Blackwell incomparability concerns V^* , and a fixed sub-optimal δ_M may fail to realize the optimal-value crossover on $\mathcal{D}$. It holds under the value-attainment hypothesis (Assumption 1).*

Remark 3 (What the caps buy, and what they do not). Neither hypothesis of (a), separation or the caps, can be dropped *from this criterion*. They are not preconditions of structural incomparability as such: for the sources this paper measures, Lemma 1 already separates W_1 from W_2 by coordinate non-refinement, with no caps at all. Lemma 2(a) is a different route to the same conclusion *within the same projection setting*: it argues from Bayes values rather than from σ -algebras, and the caps are what that argument costs. It is not a generalization to arbitrary sources; a criterion for those would be a separate statement we do not prove. What the caps do here is show why a PASS_M crossover alone proves nothing: (W_1, W_{1+}) is ($M, \mathcal{D}$)-incomparable on the measured cells (W_1 wins the API cells, W_{1+} the encoding cell) yet strictly Blackwell-comparable by construction ($W_{1+} \succeq_B W_1$, the superset revealing one further independent non-degenerate coordinate), precisely because the deprived-side premise fails there: W_{1+} contains W_1 's coordinate, so the API-side cap is not small. A bare PASS_M crossover, with neither premise verified, certifies nothing structural in *either* direction. It is equally consistent with genuine incomparability under optimal play (two

sources revealing independent coordinates cross even when each side plays optimally) and with a merely incoherent rule (Theorem 3). The two cases should not be run together. (W_1, W_{1+}) is a *strict comparability* whose PASS_M preferences nonetheless oppose; the distinct phenomenon of a rule preferring differently between two *equivalent* experiments is illustrated by the padding pair (W_1, W_{pad}) , where the added block carries no coordinate any verifier reads (Remark 2). That a representation-sensitive decoder *can* do this in general is a structural observation, not something our single measured difference certifies. Nor is the projection construction *necessary* for incomparability: it is the cleanest sufficient condition we know, and the one our sources realize.

Assumption 1 (Realized cross-advantage on the witnessing task pair). Let (T_a, T_b) be an *ordered pair of named tasks*. Say δ_M *attains value* on (T_a, T_b) with margin $\gamma > 0$ if $\text{PASS}_M(W_1, T_a) - \text{PASS}_M(W_2, T_a) \geq \gamma$ and $\text{PASS}_M(W_2, T_b) - \text{PASS}_M(W_1, T_b) \geq \gamma$; i.e. the fixed model actually exploits each source’s private coordinate on the task that requires it. Despite the historical name we keep for the label, the condition is a statement about *realized* pass rates only: it asserts nothing about the supremum in (3) being attained, and no design can compel a model to satisfy it. The hypothesis is always asserted of an explicitly named pair: it does *not* assert that some such pair exists somewhere in $\mathcal{D}$, and attainment on one pair implies nothing about attainment on another. If δ_M attains value on a pair with $\{T_a, T_b\} \subseteq \mathcal{D}$, then W_1 and W_2 are $(M, \mathcal{D})$ -incomparable.

The three objects do different jobs. Lemma 1 establishes structural incomparability for the specified projection sources, with no prior, no caps and no measurement; Lemma 2 is a *conditional* Bayes-value separation criterion whose guessing caps are assumptions, never estimates recovered from the observed state-conditional PASS_M rates; Assumption 1 concerns realized cross-advantage on named tasks, which the experiments assess directly.

5 The incomparability theorem

Section 4 showed the order is partial; we now establish the consequence that motivates the whole paper: no scalar can rank a Blackwell-incomparable pair. We *split* the claim: a theorem for query-*independent* source scores $\varphi(W)$, and a separate, explicitly empirical statement for query-*conditioned* scores $\varphi(W, T)$, the relevance score practitioners actually use; the two must not be conflated.

5.1 Query-independent scores: the theorem

Theorem 2 (No query-independent scalar ranks incomparable sources, C1). *Let $\mathcal{S}$ be finite and let W_1, W_2 be Blackwell-incomparable. (Finiteness is used only to invoke the converse direction of Theorem 1, which turns incomparability into the existence of the witness tasks; it holds for our coordinate construction, and for projection sources the witnesses are exhibited directly in the proof of Lemma 2, so no converse is needed there.) Let $\varphi : \{\text{sources}\} \rightarrow \mathbb{R}$ be any query-independent scalar score (a single number per source: relevance to a fixed corpus, mutual information $I(S; W)$, $\mathcal{V}$ -usable information, perplexity-gap, expected information gain under a fixed prior). Being real-valued, φ totally preorders sources; it need not order them strictly, since $\varphi(W_1) = \varphi(W_2)$ is permitted. Call φ faithful on a task T if the source strictly preferred on T receives the strictly larger φ -value, so a tie is a failure; a task-independent tie-breaking rule fixed in advance reduces ties to the strict case, leaving the conclusion unchanged. Then there exist tasks T_a, T_b such that φ is unfaithful to the optimal-value order on at least one of them, the witnesses being bounded-utility decision problems from Theorem 1(ii), not necessarily binary verifiers (Definition 2), though they can be taken binary for projection sources (proof of Lemma 2(a)). If moreover δ_M attains value on that same pair (T_a, T_b) (Assumption 1), φ is likewise unfaithful to the realized- PASS_M order on at least one of*

T_a, T_b . More generally, attainment on any named pair (T'_a, T'_b) yields PASS_M -disagreement on at least one of T'_a, T'_b by the same argument; the two conclusions concern their own witness pairs and do not transfer between them.

Remark 4 (Scope). Theorem 2 rules out exactly the class of *source-level* scalars: any φ that is a fixed number per source, regardless of task. $\mathcal{V}$ -usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1).

Proposition 1 ($\mathcal{V}$ -usable information is Blackwell-monotone). *Let $\mathcal{V}$ -usable information $I_{\mathcal{V}}(W \rightarrow S)$ be defined, after Ethayarajh et al. [2], as the reduction in expected predictive risk of the best predictor in a fixed family $\mathcal{V}$ when given the signal w versus the null signal. If $W_1 \succeq_B W_2$ and the predictor family $\mathcal{V}$ is closed under pre-composition with measurable garblings, then $I_{\mathcal{V}}(W_1 \rightarrow S) \geq I_{\mathcal{V}}(W_2 \rightarrow S)$. Hence $I_{\mathcal{V}}$ is a Blackwell-monotone scalar: it respects $\succeq_B$ but collapses it to one real number.*

Remark 5 (Scope of Proposition 1). The closure hypothesis (a family closed under pre-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour, $\mathcal{V}$ -usable information. For mutual information and fixed-prior EIG the analogue is not a conjecture but immediate from data processing, the direction Zheng et al. [53] use (see Section 2). A *fixed* decoder’s perplexity-gap is the singleton-family case the previous sentence excludes, so we claim nothing for it. Theorem 2 excludes all of them as per-source scalars regardless, which is the only property this paper needs.

5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score $\varphi(W, T)$, e.g. the lexical or embedding relevance that retrieval and RAG actually deploy, induces a *different* order per task and is therefore *not* a single fixed total order, so Theorem 2 does *not* cover it. We make two precise statements instead.

Proposition 2 (Constrained impossibility for feature-measurable scores). *Suppose $\varphi(W, T) = g(\psi(W, T))$ for a fixed feature map ψ that is topical: ψ is invariant to the verifier-decisive coordinate of S_T . Let T_c be a task whose decisive coordinate is supplied by W_1 , so realized utility ranks $W_1 \succ W_2$ on T_c . Parts (a) and (b) are alternatives, not a chain: their hypotheses are mutually exclusive. They are not exhaustive of the topical class, and the conclusion below is stated only for scores meeting one of them. (a) **(Tie.)** If $\psi(W_1, T_c) = \psi(W_2, T_c)$ then $\varphi(W_1, T_c) = \varphi(W_2, T_c)$, so φ cannot rank W_1 strictly above W_2 . (b) **(Strict inversion.)** Suppose instead that $\varphi(\cdot, T_c) = g(\text{ov}(\cdot, T_c))$ reads the pair through the scalar lexical-overlap feature ov alone, with g strictly increasing, and that T_c ’s lexicon is saturated with W_2 ’s vocabulary in the precise sense $\text{ov}(W_2, T_c) > \text{ov}(W_1, T_c)$. Then $\varphi(W_2, T_c) > \varphi(W_1, T_c)$, a strict mis-rank. In either case φ fails to rank $\{W_1, W_2\}$ by realized utility on T_c : no topical score satisfying (a) or (b) ranks $\{W_1, W_2\}$ correctly on all tasks.*

Remark 6 (What Proposition 2 does not exclude). Invariance to the decisive coordinate’s *value* is a weak hypothesis, and hypotheses (a) and (b) do not cover every score satisfying it. A rule reading public metadata can escape while never touching a private value. Let $S = (X, Y)$ with X, Y independent, let W_{api} carry the public label **api** and reveal X , let W_{wire} carry the public label **wire** and reveal Y , and let T_{api} and T_{wire} need X and Y . The score $\varphi(W, T) = \mathbf{1}[\text{label}(W) = \text{family}(T)]$ then ranks correctly on both tasks in every state, and it satisfies neither (a) (the labels differ, so no tie) nor (b) (it is not an increasing function of lexical overlap). So the proposition constrains the two named mechanisms, not the whole query-conditioned class, and in particular it says nothing against

routers built on public task metadata. Theorem 2, which concerns one fixed number per source for all tasks, is unaffected: a router of this kind is not a per-source scalar. What the trap shows empirically is narrower: the text-reading scorers we ran put W_2 above W_1 on it (see Section 7.4.2). That is a measured mis-rank, not evidence that a learned checkpoint satisfies (a) or (b); only the lexical cosine meets (b) by construction.

Remark 7 (Why (b) is stated for a scalar overlap score). Strict monotonicity in one *coordinate* of a multi-feature ψ does not give (b). Take $\psi = (\text{ov}, \text{style})$ and $\varphi = \text{ov} + 10\text{style}$, which is strictly increasing in overlap. With $\psi(W_1, T_c) = (0.2, 1)$ and $\psi(W_2, T_c) = (0.9, 0)$ the scores are 10.2 and 0.9: overlap favours W_2 while the score favours W_1 . Comparing two inputs that differ in other features therefore needs the whole map, which is why (b) assumes φ reads overlap alone. The lexical cosine of Section 7 satisfies that assumption by construction; a learned reranker need not, which is why its behaviour is settled empirically rather than by this proposition.

Because Proposition 2 binds a query-conditioned score only when it meets hypothesis (a) or (b) (Remark 6), we treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use and for the checkpoints of Section 7.4.2. In Section 7 the `trap_store_wire` task realizes T_c , where the lexical cosine and representative open-weight learned rankers—two dense bi-encoders (BGE, E5) and a cross-encoder reranker—all mis-rank the pair (Section 7.4.2, Table 3), so the topical class is not a lexical strawman. The same failure is documented in the wild: language-model re-rankers are steered by lexical similarity across NQ, LitQA2, and DRUID [6], a corpus of real retrieved contexts [66]; and semantically close but logically irrelevant passages defeat retrievers and rerankers at scale, with a listwise evidence-reasoning agent as the remedy [67]. An LLM *listwise* reranker, which reads both documents and can therefore condition on the private coordinate itself, ranks the trap correctly (100/100), behaviour Proposition 2 permits rather than predicts.

6 The decision-restricted deficiency surrogate and its uniform verification

The impossibility of Section 5 concerns the idealized user of a source; every quantity we report, by contrast, comes from a single fixed model. This section builds the instrument that connects the two: a surrogate that measures the order on the fixed rule, and a distribution-free verification that makes each comparison auditable on a live model.

6.1 The surrogate and its correct name

Exact garblings over text cannot be computed, and $\succeq_B$ is only partial, so instead we measure the fixed-rule order of Definition 5. Fix the model rule δ_M and a finite task set $\mathcal{D}$, and define the *decision-restricted value-deficiency*, the worst amount by which W_1 trails W_2 across the tasks in $\mathcal{D}$:

$$d_{M,\mathcal{D}}(W_1, W_2) := \max\left(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)]\right), \quad (5)$$

where each $\text{PASS}_M(W, T)$ is the *population* pass probability $p_{W,T}$ of (2), so (5) is a population functional and not a statistic; its *plug-in* estimate $\hat{\delta}_{\mathcal{D}}$ substitutes $\hat{p}_{W,T} = k_{W,T}/n$ from n completions per cell, the hat marking the sample quantity throughout. In population terms $d_{M,\mathcal{D}}(W_1, W_2) = 0$ is *equivalent* to the dominance $W_1 \succeq_{M,\mathcal{D}} W_2$ of Definition 5, and mutual positivity $d_{M,\mathcal{D}}(W_1, W_2) > 0 \wedge d_{M,\mathcal{D}}(W_2, W_1) > 0$ to $(M, \mathcal{D})$ -incomparability. Neither holds of the plug-in: at finite n , $\hat{\delta}_{\mathcal{D}} = 0$

certifies no dominance and $\widehat{\delta}_{\mathcal{D}} > 0$ no positive population gap, both carrying sampling error and the maximum being selected besides. Section 6.2 supplies the missing confidence statement, and that construction, not the sign of the plug-in, is the inferential step on which every claim we call *verified* rests.

Remark 8 (This is *not* Le Cam deficiency). The Le Cam deficiency of $\mathcal{E}$ relative to $\mathcal{F}$, experiments on a common state space $\mathcal{S}$ with kernels $\kappa_{\mathcal{E}}(\cdot | s)$, $\kappa_{\mathcal{F}}(\cdot | s)$, is the *directed* quantity

$$\delta(\mathcal{E}, \mathcal{F}) := \inf_G \sup_{s \in \mathcal{S}} \|G \kappa_{\mathcal{E}}(\cdot | s) - \kappa_{\mathcal{F}}(\cdot | s)\|_{\text{TV}}, \quad \|P - Q\|_{\text{TV}} := \sup_A |P(A) - Q(A)|,$$

with G ranging over Markov kernels from $\mathcal{E}$'s signal space to $\mathcal{F}$'s, so that $\delta(\mathcal{E}, \mathcal{F}) = 0$ recovers the garbling relation $\mathcal{E} \succeq_B \mathcal{F}$ of Definition 4 [63, 68, 69]. Its value depends on the direction, on the normalization above (the supremum over events; the L_1 form is twice it), and on $\sup_s$ rather than an average against a prior on $\mathcal{S}$, so we import no constants from the decision-theoretic reading of van Rooyen and Williamson [57], whose uniform excess-risk bound over a *normalized* bounded-loss class is tied to that normalization and unavailable under unrestricted loss scaling. The quantity in (5) performs no garbling minimization and uses a single fixed rule δ_M over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates $d_{M,\mathcal{D}}$ to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (5) the *decision-restricted value-deficiency* and use “Le Cam deficiency” only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative: $d_{M,\mathcal{D}}(W_1, W_2) = 0$ realizes the for-all-tasks-in- $\mathcal{D}$ dominance of Definition 5, the measurable shadow of Corollary 1.

6.2 The uniform verification

The surrogate (5) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

Proposition 3 (Uniform incomparability verification). *Within each cell (W, T) draw n i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let $[L_{W,T}, U_{W,T}]$ be the exact (two-sided) Clopper–Pearson interval [9] for $p_{W,T} = \text{PASS}_M(W, T)$ at level $\alpha = \eta/(2|\mathcal{D}|)$. By the Bonferroni/union bound [10] over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T \in \mathcal{D}}$, all $2|\mathcal{D}|$ intervals hold simultaneously with probability at least $1 - \eta$. On that event,*

$$L_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [L_{W_2,T} - U_{W_1,T}] \tag{6}$$

*is a $(1 - \eta)$ lower confidence bound on $\sup_{T \in \mathcal{D}} [p_{W_2,T} - p_{W_1,T}]$. Hence $L_{\mathcal{D}}(W_1, W_2) > 0$ **verifies** the population statement $d_{M,\mathcal{D}}(W_1, W_2) > 0$ at confidence $\geq 1 - \eta$. Both directions $L_{\mathcal{D}}(W_1, W_2)$ and $L_{\mathcal{D}}(W_2, W_1)$ read off the same $2|\mathcal{D}|$ intervals and so are valid on the same coverage event. Consequently, if both observed bounds are positive, the joint incomparability claim $\{d_{M,\mathcal{D}}(W_1, W_2) > 0 \wedge d_{M,\mathcal{D}}(W_2, W_1) > 0\}$ holds at $\geq 1 - \eta$ with no additional correction. The condition is not decoration: coverage alone implies nothing about the population gaps, which are identically 0 when W_1 and W_2 are the same source. The verification is one-sided, asserting nothing when the two bounds are not both positive, and the guarantee is frequentist: the probability of declaring incomparability when it does not hold is at most η , which is not a posterior probability that the fixed sources satisfy the inequality. The matching quantity $U_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [U_{W_2,T} - L_{W_1,T}]$ is, on the same event, a $(1 - \eta)$ upper confidence bound on the same supremum; the dominance control uses it to verify $d_{M,\mathcal{D}}(W_1, W_2) = 0$ up to the equivalence margin.*

A naive single-sample DKW bound under-covers this supremum, because the object is a maximum of differences of binomial means across distinct cells and the verification selects the argmax-gap cell, so the multiplicity must be paid explicitly rather than absorbed into a single-sample envelope (Appendix A); the resulting construction aligns with recent statistical guidance for black-box LLM evaluation (Appendix A).

Assumption 2 (Sampling regime). The n outcomes within a cell are i.i.d. Bernoulli draws of $v_T(\delta_M(T, w))$. Every draw is a single completion (plus deterministic code extraction) from an independent call with no shared state, which is what makes the binomial model of Assumption 2 reasonable rather than proved; we report the decoding configuration in Section 7. Where draws are near-deterministic ($\hat{p} \in \{0, 1\}$ stark cells) the Clopper–Pearson interval becomes one-sided but retains nonzero width, so the verification stays conservative rather than degenerate.

6.3 The interference verification (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source W_{1+} (revealing strictly more coordinates than W_1 ; Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T := \text{PASS}_M(W_{1+}, T) - \text{PASS}_M(W_1, T) - \text{PASS}_M(W_2, T) + \text{PASS}_M(\text{none}, T), \quad (7)$$

the second-difference (interaction) of adding W_2 on top of W_1 . We write the interaction as Ψ_T (rather than a symbol I that would collide with mutual information $I(S; W)$). A verified $\Delta_T < 0$ on a cell whose W_1 arm is at ceiling is the localized statement that “more strictly-more-informative context hurts”; Ψ_T is reported alongside it as the interaction term only (Remark 9).

Proposition 4 (Interference verification). *Fix a target task $T^* \in \mathcal{D}$ chosen before unblinding, and let $\Delta_{T^*} := \text{PASS}_M(W_{1+}, T^*) - \text{PASS}_M(W_1, T^*)$ be the harm contrast. With exact Clopper–Pearson endpoints $\text{CP}_{\text{lo}}, \text{CP}_{\text{hi}}$ carrying noncoverage at most $\alpha = \eta/2$ per endpoint on the two arms $\{W_{1+}, W_1\}$ (two endpoints, union-bounding to η ; the two unused endpoints of a two-sided interval are not paid for),*

$$\bar{\Delta}_{T^*} := \text{CP}_{\text{hi}}(p_{W_{1+}}) - \text{CP}_{\text{lo}}(p_{W_1}) \quad (8)$$

is a $(1 - \eta)$ upper confidence bound on Δ_{T^} , and $\bar{\Delta}_{T^*} \leq -\tau$ verifies $\Delta_{T^*} \leq -\tau$ at confidence $\geq 1 - \eta$. If instead the harmful cell is selected from $|\mathcal{D}|$ candidates after unblinding, use $\alpha = \eta/(2|\mathcal{D}|)$; this is the same selection correction the verification applies to the deficiency surrogate (the multiplicity is real: the selected cell is an argmax over $|\mathcal{D}|$ noisy gaps, so a single-sample sup inequality does not cover it), and omitting it would repeat the error we criticize. Because Δ reads only the two arms it contrasts, the bound is insensitive to the **none** baseline; we still report that baseline per cell, because guessability bears on whether the source is genuinely private and hence on the incomparability reading (see Section 3.3).*

Remark 9 (Ψ is interaction, not harm). The second difference must not be read as a harm measure. Writing f for the set function $A \mapsto \text{PASS}_M(W_A, T)$, the inequality $\Psi_T \leq 0$ is precisely the submodularity condition for the pair $\{W_1, W_2\}$, so a negative Ψ_T is what diminishing returns look like and is consistent with f being monotone submodular. It does not imply that adding W_2 to W_1 hurt: taking $\text{PASS}(\text{none}) = 0$ and $\text{PASS}(W_1) = \text{PASS}(W_2) = \text{PASS}(W_{1+}) = 1$ gives $\Psi_T = -1$ with no degradation whatever. What the collapse violates is *monotonicity*, $\Delta_T < 0$, which is the quantity we verify. In our data the two coincide in direction, and the contrast bound is tighter than the four-term one because it spends its confidence budget on two intervals rather than four.

6.4 Anti-monotonicity as information non-monotonicity of δ_M

A verified $\Delta_{T^*} < 0$ on a ceiling- W_1 cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

Theorem 3 (Information non-monotonicity of the model rule). *Let W_{1+} reveal a superset of W_1 ’s coordinates, so $W_{1+} \succeq_B W_1$ (Lemma 1). Call a rule δ Blackwell-coherent on T if its realized pass rate does not fall when it is handed the dominating source, that is $\text{PASS}_\delta(W_{1+}, T) \geq \text{PASS}_\delta(W_1, T)$ at the realized state S_T . Coherence in this sense costs no information, and a coherent rule exists on every task: the pruning rule $\delta_M \circ G$, where G deletes W_2 ’s block from W_{1+} ’s signal and acts as the identity on W_1 ’s, so that the one rule is defined on both inputs, satisfies $\text{PASS}_{\delta_M \circ G}(W_{1+}, T) = \text{PASS}_{\delta_M \circ G}(W_1, T) = \text{PASS}_M(W_1, T)$ at that state, because $G \circ \kappa_{W_{1+}}(\cdot | s) = \kappa_{W_1}(\cdot | s)$ for every s ; no prior and no averaging enter. The Bayes rule δ^* is deliberately not the exemplar here: its value is monotone in the μ_T -averaged register only, and monotonicity there is consistent with a fall at a single state (Remark 1). If a fixed model rule δ_M has a task T^* with $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$, then δ_M is not Blackwell-coherent: it violates monotonicity under the garbling order on T^* . In particular no monotone scalar context-value score can predict δ_M ’s behaviour on T^* .*

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent in the PASS register, and the comparison is like-for-like: δ_M is beaten, at the same state and on the same input, by its own pruned version $\delta_M \circ G$. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.

6.5 Family-wise error across the verified claims

Each verification above controls error *marginally* at η for its own claim (incomparability via Proposition 3; interference via Proposition 4). The unit of accounting is the *coverage event*, not the claim. Every statement that is a deterministic consequence of the same $2|\mathcal{D}|$ Clopper–Pearson intervals holds simultaneously on the event E of Proposition 3, and therefore *jointly* at $\geq 1 - \eta$ with no further correction: this covers both incomparability directions, the dominance control $U_{\mathcal{D}}(\cdot, \cdot)$, whose endpoints are endpoints of those very intervals, and any per-task strict win read off them. Adding a conclusion drawn from intervals already covered does not by itself cost another Bonferroni factor. What does require fresh allocation is a conjunction spanning *different* events, that is, claims reading cells outside E or intervals built at a different level. The interference verification is of that kind, since it reads the W_{1+} and W_1 arms at $\alpha = \eta/2$ per endpoint, and the W_{1+} intervals are not in the W_1/W_2 event at all; conjunctions joining it to the incomparability family are therefore reported as marginal per claim, and where a joint statement is wanted we allocate η across the C events involved.

The two verifications have different effective- n requirements: incomparability is carried by stark 0/1 cells whose intervals are tight, whereas a harm contrast on a milder collapse spends more of the budget. This is why the four-term interaction bound left Opus short of the margin while the two-arm contrast clears it (see Section 7.4.3); the power analysis is in Appendix A.

7 Experiments

With the order (see Section 4), its impossibility consequence (see Section 5), and the verification that measures it (see Section 6) in hand, we now instantiate the theory on real language models. The

design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (see Section 7.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every verified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (see Section 7.4), so the verification is auditable rather than asserted.

7.1 Design: two incomparable context sources

We define two *private*, *unguessable* conventions on the *same* module `ledger`, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module, same task shape). Because both are on-topic for any “ledger” task, any lexical, embedding, mutual-information, or $\mathcal{V}$ -usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

W_1 , **API call contract**. `ledger.post(account, amount, *, memo)`: `account` first, `amount` second positional, `memo` a *required keyword-only* argument; returns an integer entry id ≥ 1 ; raises `ledger.PostError` (not `ValueError/KeyError`) on failure.

W_2 , **wire-encoding invariant**. Amounts serialize to `<sign><minor-unit-digits>#<check>`, where `check` = (sum of digits) mod 10; e.g. `$12.30` $\mapsto$ `" +1230#6"` and `-$0.07` $\mapsto$ `" -7#7"`. Decoding rejects a wrong check digit.

W_{1+} , **literal superset**. $W_{1+} = W_2 ++ W_1$, the concatenation of W_2 ’s signal with W_1 ’s. As a deterministic source revealing both coordinates, it Blackwell-dominates both W_1 and W_2 ($W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$); it is the anti-monotonicity / dominance probe. We also use $W_{1+}^{\text{rev}} = W_1 ++ W_2$ (W_1 first) as an order control.

Why incomparable (garbling argument). The argument is simple. From W_2 ’s wire format you cannot recover W_1 ’s private argument order, its required `memo` keyword, or its error type; from W_1 ’s contract you cannot recover W_2 ’s cents scale, its `#` separator, or its mod-10 check. Each source states a fact the other leaves out, so neither is a blurred copy of the other, and W_1, W_2 are incomparable. This is exactly the coordinate picture of Section 3: W_1 and W_2 reveal different, non-overlapping facts about S (Lemma 1).

Confound defense: the double crossover. The key pattern is a *double crossover*, measured against a **none** baseline (no note supplied): W_1 raises the API tasks but *not* the encoding task, while W_2 raises the encoding task but *not* the API tasks. If encoding were simply “harder,” W_1 would fail to help it and W_2 would still have to raise it from the same floor. In the clean carrying cells the deprived arms floor at or near the **none** baseline, so the two task families are equally hard on their own and any difference comes from the matching private fact, not from one family being harder; when a model prior leaks a convention (Table A1), we report that leakage explicitly and do not use those cells for claims that require a near-zero baseline.

7.2 Tasks

The full design comprises seven tasks with stub-`ledger` verifiers (return code 0 = PASS): four W_1 -decisive API tasks (`api_post_ok`, `api_argorder`, `api_error_type`, `api_memo_required`), two

W_2 -decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

Verified battery. The verified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the verified battery. The four-cell selection was made to retain at least one W_1 -decisive, one W_2 -decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (see Section 7.4.1).

The relevance trap. The trap `trap_store_wire` is a constructed test of the topicality condition in Proposition 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks $W_2 > W_1$; but its graded requirement is the W_1 call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks $W_1 > W_2$. This defeats the query-conditioned relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine $\varphi_{\text{lex}}(W, T)$ between a source and the task prompt, computed *before* the model is run. We use lexical cosine as the *minimal* instantiation of Proposition 2, the cheapest member of the topical class; stronger learned rankers (dense bi-encoders, a cross-encoder, and an LLM listwise reranker) are tested against the same trap separately in Section 7.4.2, while the theorem (Theorem 2) is the general claim for query-independent scalars.

7.3 Materials and Methods

Models and vendors. We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6. Nothing in this paper is trained or fine-tuned: all six are fixed production systems accessed through their public APIs, so there are no training or validation curves to report; the released measurement record, per-cell PASS tables, exact intervals, and raw run logs, is the complete empirical object.

Harness and arms. The harness (`harness/measure_blackwell.py`; released with all run logs at <https://github.com/abilous-ti/blackwell-llm-context>) is Python-standard-library only (zero external dependencies); it computes $\hat{\delta}_{\mathcal{D}}$ in both directions, the uniform Clopper–Pearson/Bonferroni verification, the interference verification, and the lexical relevance score, and it implements the launchers for the Anthropic Messages API, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms $\{\text{none}, W_1, W_2, W_{1+}\}$. PASS is measured over n completions per cell.

One turn budget, one transport. Every model gets a single completion per attempt, from which we extract the candidate code and grade it. All six are reached the same way: a single HTTP request the harness composes itself, carrying only the task prompt, with no system prompt of our own and no tools available to the model. The Anthropic models are served over the Messages API and the other three over an OpenAI-style endpoint, but in both cases the request is one pass with no agentic exploration, so the model must use what it is given. Cross-vendor *magnitudes* therefore no longer carry a transport confound, though they remain subject to the ordinary caution that

vendors’ models differ in ways we do not control (see Section 8.4). The claims the paper verifies are within-model comparisons between arms, where any remaining request-level detail is held fixed and cancels.

Statistical reporting. Each cell is $n = 40$ independent completions scored by an executable verifier and summarised as a count; the arms within a cell are separate completions, and each draw is an independent call with no shared state (Assumption 2). Uncertainty is carried by exact Clopper–Pearson intervals rather than a normal approximation, which at $n = 40$ and proportions near 0 or 1 is where the approximation is worst (see Section A). Multiplicity is paid explicitly by a union bound whose family is stated with each claim, at confidence budget $\eta = 0.10$, so the statements *within* a family hold jointly with probability $\geq 1 - \eta$. Incomparability spends $\alpha = \eta/(2|\mathcal{D}|)$ over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T \in \mathcal{D}}$, and both directions are read off those same intervals (Proposition 3); a harm contrast spends $\alpha = \eta/2$ on its two arms $\{W_{1+}, W_1\}$ and verifies $\Delta_T \leq -\tau$ at $\tau = 0.30$ through the upper bound $\bar{\Delta}_T$ (Proposition 4), the four-term Ψ_T being reported descriptively only (Remark 9). Each verification is thus controlled marginally at η ; we do not claim joint control across the incomparability, dominance and interference families drawn from a single run (see Sections 6.5 and 8.4). At $\hat{p} = 0$ or $\hat{p} = 1$ the interval becomes one-sided but does not degenerate to a point: at $n = 40$ and the level Figure 1 uses, $0/40$ gives $[0, 0.1281]$ and $40/40$ gives $[0.8719, 1]$. It is that residual width a crossing must beat, which is why the stark 100%/0% cells verify at this n while intermediate proportions would need a larger one. Results whose intervals do not separate are labeled *reproduced but not verified* and are never used as evidence; the protocol reports its own insufficiency rather than hiding it. We report no p -values and fit no model to the data.

Reproducibility artifacts. The measurement harness, every per-cell run log, every result file and the analysis scripts are released (Data Availability Statement). `results/MANIFEST.md` pins each record by SHA-256 and maps every table and figure to the files it is computed from, and `harness/diag/check_regimes.py` reports the transport and turn budget of each cited run with the evidence for it. It is an inventory, not a proof: it checks only the runs on a fixed list, and matches the disclosure wording against the manuscript globally rather than to the run it describes. Appendix D gives the decoding configuration, the retry and outage policy, and the cells that were re-measured, with why.

Why these constants. Both constants were fixed before the runs and neither is load-bearing. $\eta = 0.10$ is a conventional 90% family level, and the headline separations are near-0% against near-100%, so tightening it does not move the verified conclusions. $\tau = 0.30$ is an a priori materiality threshold: to count, interference must destroy at least 30 PASS points, well above the interval widths at $n = 40$, so noise cannot clear it. The data show the threshold doing real work. Verified collapses clear it with upper bounds from -0.38 to -0.85 (Table 4), and the superset falls below the character-matched padding arm at -0.85 on both control cells. What shows the choice is not load-bearing is that the verdict barely moves with it: every one of the twelve contrasts clears marginally for any τ from 0.10 to 0.30, and eleven of twelve up to 0.50, so halving or raising the threshold leaves the conclusion intact. The margin is set by DeepSeek’s two cells, at -0.384 and -0.540 , which are the first to drop out as τ rises.

What the intervals do not cover. The Clopper–Pearson intervals carry sampling error under Assumption 2, a fixed success probability per cell. They carry nothing else, and two components

Table 2: Consolidated 6-model / 4-vendor verdict. Every model gets one completion per attempt at $n = 40$ per cell, and all six models are queried over the same HTTP path (see Section 7.3), so what separates vendors here is the models, not the request path. “Anti-mono. verified” counts, of the two API cells tested per model, those on which the harm contrast’s upper bound clears $-\tau = -0.30$ at $\eta = 0.10$. “Argorder none” is the no-context baseline PASS on `api_argorder`, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers. Confidence bounds are rounded away from the claim they support—a lower bound down, an upper bound up—so every printed bound is implied by the computed one.

Claim	Haiku	Sonnet	Opus	GPT-5.5	DeepSeek	Kimi-K2.6
Vendor	Anthropic	Anthropic	Anthropic	OpenAI	DeepSeek	Moonshot
n per cell	40	40	40	40	40	40
Incomp. verified ($\geq 90\%$)	✓	✓	✓	✓	✓	✓
$L_D(W_1, W_2)$	+0.22	+0.76	+0.76	+0.76	+0.34	+0.67
$L_D(W_2, W_1)$	+0.76	+0.76	+0.76	+0.76	+0.71	+0.76
Relevance mis-rank (trap)	✓	✓	✓	✓	✓	✓
Anti-mono. verified (# cells, of 2)	2	2	2	2	2	2
<code>api_argorder none</code>	0%	0%	100%	75%	2%	45%

outside them are larger than they are. *Model drift*: successive re-measurements moved Opus’s `api_post_ok` superset arm from 38% to 12% and then to 3%, and DeepSeek’s W_1 arm from 75% to 43%, both far outside the nominal intervals for those cells. *Prompt wording*: an earlier record traced a 100% $\rightarrow$ 52% swing on one cell to the prompt naming the call as `ledger.post`, an effect an order of magnitude wider than any interval here, and each cell has exactly one prompt, so this component is not modelled at all. Every verified quantity is therefore an inference about a (model snapshot, prompt, default decoder, collection window) tuple, indexed by θ (see Section 3.3) and by the run it was drawn in, not a property of a vendor’s model line. We read that as a weakness: the re-measurement bounds the drift component empirically, and the direction of every verdict survived it.

Provenance. Every cell of the six-model grid is read from that model’s own complete run, so each harm contrast compares two arms collected in one window; the same holds for the four behavioural controls, which share a single run, and for the second source pair. One substitution remains: both the W_1 and the W_2 conditions of GPT-5.5’s `trap_store_wire` task are taken from the retained, transport-clean re-measurement, at 40/40 and 0/40, because the base run logged an HTTP 500 there. Those two therefore share a window with each other; what crosses windows is that task against GPT-5.5’s other cells, where a shared model name and transport do not establish a shared decoder (Appendix D). Raw completions were retained for three of the six models (Appendix C.1); for the other three the record is aggregate-only. `results/MANIFEST.md` gives the run, and so the window, behind each number.

7.4 Results

We report four findings: incomparability (see Section 7.4.1), the relevance mis-rank (see Section 7.4.2), anti-monotonicity (see Section 7.4.3), and the model-relative value of private context (see Section 7.4.4). Table 2 consolidates the per-model verdicts; Tables 4 and A1 give the underlying quantities.

7.4.1 Incomparability verified on all six models

On every model the empirical deficiency is mutually positive, $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ and $\hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$, and the uniform lower bounds $L_{\mathcal{D}}$ in both directions clear zero, verifying empirical incomparability at $\geq 90\%$ (Table 2). The crossover is stark: W_1 solves the API and trap cells and fails encoding, while W_2 solves encoding and fails the API and trap cells. For Haiku at $n = 40$ the two directions are attained by different cells: $L_{\mathcal{D}}(W_2, W_1) = +0.762$ by `api_post_ok`, where W_1 passes 40/40 and W_2 none, and $L_{\mathcal{D}}(W_1, W_2) = +0.226$ by `enc_amount`, where W_2 passes 22/40 and W_1 none. The encoding contrast is the binding direction, narrower because W_2 reaches only 22/40 there.

The six-model claim, made jointly. Proposition 3 controls the $2|\mathcal{D}|$ intervals of *one* model’s run, so the per-model verifications above are each at $\geq 1 - \eta$ and their conjunction across the roster is not. Rather than leave the six-model statement marginal we pay for it: allocating η across the six runs, i.e. per-*interval* level $\eta/(2|\mathcal{D}| \cdot 6) = \eta/48$, which places $\eta/96$ in each tail, every one of the twelve directions remains strictly positive, binding at Haiku’s $L_{\mathcal{D}}(W_1, W_2) = +0.146$ and reaching $+0.684$ on the stark cells. The claim “all six models are empirically incomparable” therefore holds *simultaneously* at $\geq 90\%$, not merely model by model. Table 2 reports the per-model bounds; the simultaneous ones are uniformly smaller by 0.077 to 0.084 and change no verdict.

Because incomparability is a two-directional claim, both $L_{\mathcal{D}}$ bounds must be positive; the smaller of the two is the binding one. Two cells carry each model’s verification, and which two is model-specific: on five models they are one API cell and the encoding cell, while on DeepSeek-V4-Pro the W_1 -over- W_2 direction is carried by the *trap* cell rather than an API cell (its API margins being smaller). The effective support of the surrogate is therefore two cells per model, drawn from a battery of four. We therefore describe $\mathcal{D}$ as a fixed adversarial four-cell battery rather than a sampled task distribution: the verification is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition run against six models—six rules on one pair, with W_1 , W_2 and $\mathcal{D}$ held fixed and only δ_M varying, not six independent source-pair replications. The n draws behind a cell are repeated outputs on *one* task, which is what the Bernoulli model of Assumption 2 prices; no interval in this paper samples tasks.

7.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$, so it prefers W_2 ; yet realized PASS ranks $W_1 \gg W_2$ on all six models (Haiku 95% vs. 0%; Sonnet, Opus, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%). On the two non-trap cells relevance *agrees* with PASS, on the API cells relevance ranks $W_1 > W_2$ and so does PASS, and on the encoding cell relevance ranks $W_2 > W_1$ and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The **none** floor on the trap is 0% on all six models, and W_2 scores 0% there too, confirming both that the convention is unguessable and that encoding knowledge is useless on this task.

The ranker spectrum is the point here: dense retrievers and a cross-encoder fall for the trap while a listwise reranker escapes it. To test whether the mis-rank extends beyond bag-of-words, we scored $\{W_1, W_2\}$ against the probe tasks with representative open-weight rankers, two dense bi-encoders (`bge-small-en-v1.5`, `e5-small-v2`) and a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`), and with a RankGPT-style LLM *listwise* reranker (Haiku one turn, seeing the task and *both* sources; both presentation orders; $n = 100$ judgments per task). Table 3 reports the induced rankings. Every feature-based ranker we tested mis-ranks the trap: both dense bi-encoders prefer W_2 on the trap while ranking both non-trap cells correctly, textbook topical behaviour, and the cross-encoder

Table 3: The ranker spectrum on the probe tasks. Each cell is the ranker’s top pick between $\{W_1, W_2\}$; “PASS” is the realized-utility ranking from the verified record. Dense and cross-encoder scores are deterministic; the LLM listwise row pools $n = 100$ judgments per task (unanimous, both presentation orders). $\times$ marks a mis-rank. The cross-encoder’s `enc_amount` pick is a near-tie (-0.279 vs -0.305).

Ranker	Family	trap (PASS: W_1)	api_post_ok (PASS: W_1)	enc_amount (PASS: W_2)
bag-of-words cosine	lexical	$W_2 \times$	W_1	W_2
bge-small-en-v1.5	dense bi-encoder	$W_2 \times$	W_1	W_2
e5-small-v2	dense bi-encoder	$W_2 \times$	W_1	W_2
ms-marco-MiniLM-L-6-v2	cross-encoder	$W_2 \times$	W_1	$W_1 \times$
Haiku listwise ($n=100$)	LLM listwise	W_1	W_1	W_2

prefers W_2 on the trap with maximal confidence (logits -5.63 for W_1 vs $+3.24$ for W_2) while being near-tied (and narrowly wrong) on the encoding cell. The LLM listwise reranker, which can read the verifier-decisive coordinate rather than match surface features, tracks realized PASS everywhere, unanimously: 100/100 per task, split evenly across the two presentation orders so that a ranker merely preferring the first document would show. Unanimity at $n = 100$ puts the exact 95% Clopper–Pearson lower bound on the preference rate at 0.964; at the pre-declared halfway checkpoint of 50 judgments it was 0.929, and 10 judgments would have bounded it only at 0.692, too close to chance for a claim about consistent ranking. This sits on the boundary Proposition 2 draws: its hypothesis is *topicality*, invariance to the decisive coordinate, and the trap defeats the lexical, dense and cross-encoder checkpoints we ran while sparing the one ranker that reads content. Two limits on how far this generalizes. The three families are three checkpoints on one trap, so their failure is measured, not proved for the families at large; and the proposition does not establish that a model call per candidate is *necessary* to escape, since a router reading public task metadata would also escape at no such cost (Remark 6). What the measurement does show is that the scorers actually deployed here, which read text, pay that price.

7.4.3 Anti-monotonicity: a verified superset that hurts

The superset W_{1+} , which literally *contains* W_1 ’s signal, sharply degrades the API cells relative to W_1 alone. The per-task harm contrast $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$ is verified negative ($\Delta_T \leq -\tau$, $\tau = 0.30$) on all twelve API cells, two per model, and so on every one of the six models (Table 4, Figure 2). Those twelve are *marginal* verifications, each at $\geq 1 - \eta$ for its own cell; the conjunction can be bought from the same records, since spreading η across all twelve contrasts, a per-interval level of $\eta/12$ over the 24 endpoints, still leaves eleven clear of $-\tau$ with every model keeping at least one. The roster-level statement “on each of these six models some strict superset verifiably harms some cell” therefore holds *simultaneously* at $\geq 90\%$, not merely model by model. The mildest verified collapse is DeepSeek’s `api_argorder`, $68\% \rightarrow 5\%$ at upper bound -0.38 , the one cell the joint budget drops; all twelve also survive a Bonferroni correction across the two cells within each model. Haiku’s shallowest collapse is on `api_argorder`, $80\% \rightarrow 0\%$, shallow only because its W_1 arm is not at ceiling. This is a verified empirical falsification of superset-*monotonicity* for these fixed rules (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it. Table A1 gives the same collapse arm by arm for every model.

The idealized Blackwell order on $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ collapses the choice to the superset: W_{1+} reveals both coordinates, so $W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$, and “more signal is better” keeps $\{W_{1+}\}$

Idealized Blackwell order (optimal user) Verified empirical order (fixed rule δ_M , Haiku $n=40$)

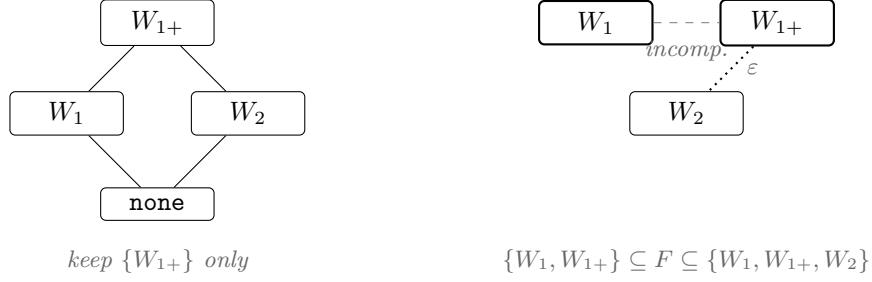


Figure 1: Two orders over the same three sources, $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ with $W_{1+} = W_2 ++ W_1$. Left: the Blackwell order for the optimal user, where the superset reveals both coordinates and dominates. Right: the order realized by the fixed model rule, on Haiku at $n = 40$, read off a single coverage event over the 3×4 Clopper–Pearson intervals (text above). A dashed edge is a verified incomparable pair; a dotted edge is an ε -dominance only, at $\varepsilon = 0.1281$, so W_2 cannot be excluded exactly. The **none** arm is drawn only on the left, since it is outside that coverage family and no relation to it is verified. The right-hand panel is therefore a verified *partial* picture rather than a determined Hasse diagram: what it establishes is $\{W_1, W_{1+}\} \subseteq F \subseteq \{W_1, W_{1+}, W_2\}$.

alone. The verified empirical relation for the fixed rule has the opposite shape (Figure 1): W_1 beats W_{1+} on the API cells, W_{1+} beats W_1 on the encoding cell, and W_1 and W_2 are the verified incomparable pair, so the *point estimate* of the PASS_M -non-dominated front is the two-element set $\{W_1, W_{1+}\}$, an object no scalar top-1 rule can represent; the two-sided inclusion below bounds it. Three sources over four cells need *one* coverage event, so all 3×4 Clopper–Pearson intervals are taken at per-interval level $\eta/12$; they hold jointly at $\geq 1 - \eta$ and every relation in the figure is read off that single event. On the $n=40$ Haiku record $L_{\mathcal{D}}(W_1, W_2) = +0.2070$ and $L_{\mathcal{D}}(W_2, W_1) = +0.7439$, both positive, which verifies the incomparable pair. Non-dominance is likewise a *lower-bound* statement, and the same event supplies it: $L_{\mathcal{D}}(W_{1+}, W_1) = +0.7439$, witnessed by `api_post_ok`, shows that W_1 beats W_{1+} on some cell, and $L_{\mathcal{D}}(W_1, W_{1+}) = +0.7439$ shows the converse, so neither dominates the other. An upper bound could carry neither claim: it bounds how large an advantage *might* be, not that one exists. The relation the front turns on has only an upper bound, $U_{\mathcal{D}}(W_{1+}, W_2) = 0.1281$. That is an ε -dominance rather than an exact one, saying that on *every* cell of the battery W_2 can beat W_{1+} by at most 0.1281, so excluding W_2 is licensed only up to that margin. What the coverage event establishes exactly is the two-sided inclusion

$$\{W_1, W_{1+}\} \subseteq F \subseteq \{W_1, W_{1+}, W_2\},$$

where F is the PASS_M -non-dominated front on $\mathcal{C}$: both W_1 and W_{1+} are verifiably in it, and at $n = 40$ a three-element front cannot be ruled out. The selection rule the figure implies, keep the non-dominated set rather than force the superset, holds under either reading.

What the position controls show. On Haiku, the collapse persists under reversal of the two source blocks, and character-matched off-topic padding does not reproduce it. The padding control (Appendix B) fills the slot W_2 occupies inside W_{1+} , W_1 last in both, and leaves the padded arm at ceiling on `api_argorder` (40/40, against W_1 alone’s 31/40): changing only that slot’s content turns a harmless arm harmful. The reversed arm $W_{1+}^{\text{rev}} = W_1 ++ W_2$ puts W_1 first, over the same transport and $n = 40$ as the main table (`blackwell_controls_api_n40`). The two orderings are

Table 4: Anti-monotonicity verification on both API cells of all six models. $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$ is the harm contrast; $\bar{\Delta}_T$ is its $\eta/2$ Clopper–Pearson upper bound over the two arms, and the cell is verified harmful when $\bar{\Delta}_T \leq -0.30$. The $W_1 \rightarrow W_{1+}$ column reports the PASS collapse. Unlike the four-term interaction Ψ_T , the contrast does not read the `none` arm, so a leaky baseline cannot slacken it.

Model	Carrying cell	$W_1 \rightarrow W_{1+}$	Δ_T	$\bar{\Delta}_T$	Verified
Haiku-4.5	<code>api_argorder</code>	80% $\rightarrow$ 0%	-0.80	-0.59	✓
Haiku-4.5	<code>api_post_ok</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
Sonnet-4.6	<code>api_post_ok</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
Sonnet-4.6	<code>api_argorder</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
GPT-5.5	<code>api_post_ok</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
GPT-5.5	<code>api_argorder</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
DeepSeek-V4-Pro	<code>api_post_ok</code>	75% $\rightarrow$ 0%	-0.75	-0.54	✓
DeepSeek-V4-Pro	<code>api_argorder</code>	68% $\rightarrow$ 5%	-0.62	-0.38	✓
Kimi-K2.6	<code>api_post_ok</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
Kimi-K2.6	<code>api_argorder</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
Opus-4.8	<code>api_post_ok</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓
Opus-4.8	<code>api_argorder</code>	100% $\rightarrow$ 0%	-1.00	-0.85	✓

indistinguishable on both cells: W_{1+} and W_{1+}^{rev} each take W_1 from 39/40 to 0/40 on `api_post_ok` and from 31/40 to 0/40 on `api_argorder`, verified harmful at -0.81 and -0.56 respectively. Moving the added block out of the primacy slot therefore changes neither the sign nor the size of the harm, which constrains a positional account of the collapse without ruling one out for configurations we did not test. On one model and in this configuration, the controls constrain positional and gross-length explanations without identifying a unique cause or isolating tokenization, processing difficulty, or the internal mechanism; the reversed arm bounds the *existence* of a positional explanation, not position’s contribution to the harm’s size.

Robust across the capability ladder. The harm is large and robust across the Anthropic capability ladder. On the two API cells the $W_1 \rightarrow W_{1+}$ collapse is Haiku 100%/80% $\rightarrow$ 0%/0%, Sonnet 100% $\rightarrow$ 0%/0%, and Opus 100% $\rightarrow$ 0%/0% (Table 4). Both stronger models sit at the floor on both cells, so this record orders neither for nor against a capability trend: an earlier two-point reading suggested the harm “intensifies with strength”, and we now claim only robustness across the ladder, not a capability trend. All three Anthropic models collapse to 0% on both cells, so there is no residual spread to order. What survives is the negative claim, not the ranking that motivated it.

7.4.4 The model-relative value of private context

The “private” conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The `none`-arm baseline on `api_argorder` forms a clean spectrum: Sonnet and Haiku 0%, DeepSeek 2%, Kimi 45%, GPT-5.5 75%. Opus is excluded from this comparison: it solves `api_argorder` unaided in 40/40 draws, so for that model the cell measures no value of private context at all (Table 2). So the value of the private context, the model-relative guessability $\text{PASS}_M(\text{none}, T)$ of Section 3.3, varies continuously across models, and it is not ordered by capability. What it does track we cannot identify from these runs. DeepSeek, a different vendor, has a tight prior like the smaller Claude models, while GPT-5.5’s prior already holds much of

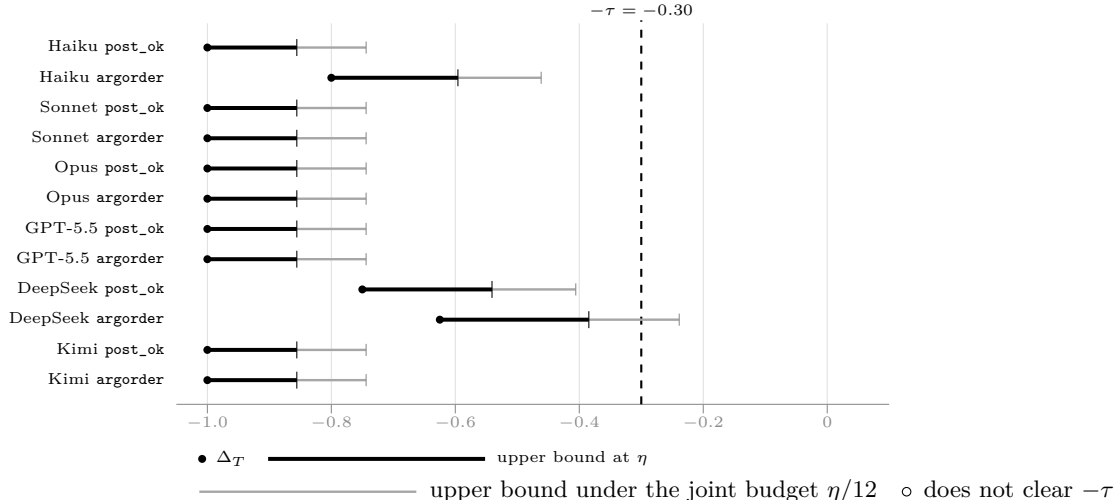


Figure 2: The harm contrast $\Delta_T = \text{PASS}_M(W_{1+}, T) - \text{PASS}_M(W_1, T)$ on both API cells of every model, with the bound that carries the verification. A cell is verified harmful when the *upper* bound clears $-\tau = -0.30$, so a bar ending left of the dashed line is a verified collapse. Thick bars are the marginal bounds at $\eta = 0.10$, one family per cell: all twelve clear. Thin grey bars are the same contrasts under a single η spread across all twelve, at per-interval level $\eta/12$: eleven clear, and every model keeps at least one. Values are those of Table 4, recomputed from the released per-cell counts.

the convention (`api_argorder` 75%, `enc_amount` 55%), so a “stronger-model-knows-more” account does not fit. Utility-centric retrieval reaches the same conclusion from the other side, the same passage carrying different utility for different models [4]; what the spectrum adds is the verified per-model floor. Within Anthropic, no-context PASS on `api_argorder` runs from 0/40 for Haiku and Sonnet to 40/40 for Opus. These observations do not identify how guessability relates to general model capability or to training data; no independent capability scale is measured here. The verified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok` `none` = 5%, `trap` = 0%).

7.4.5 What the failing replies contain

Raw completions were retained for the three Anthropic models’ rows of the interference table, 40 per arm and cell; the other three models’ rows are aggregate-only, so what follows is read off half the table (Appendix C). Three text screens bear on the collapse, over the 480 replies of the W_1 and W_{1+} arms there. The empty-extraction and explicit-refusal screens return no matches at all. The import screen detects one omission among the 240 superset replies and seven among the 240 W_1 replies, the latter concentrated in the arm that passes, so those omissions cannot explain the superset arm’s collapse. Each screen bears only on what it tests—an empty extraction, a missing import, refusal language—rather than on every packaging or declining behaviour, and none was run on the three models whose draws were not retained. *The harmful arm is verbose*: on Haiku the mean reply runs 105 characters against 477 on `api_post_ok` and 459 against 633 on `api_argorder`, the same inflation reported in Section 8.3.

What the retained text does *not* establish is the mechanism. A textual flag for W_2 ’s wire encoding appearing in the reply, stated as a rule in Appendix C.1, fires on 240/240 of the superset replies across the three models and both collapse cells and on 0/240 of the W_1 replies. It also fires on 238/240 of the W_2 replies in those cells, every one of which fails the API verifier, so it is not a

test of correctness. These counts describe an output pattern associated with the source condition; they do not identify an internal cause. Theorem 3 constrains any rule whose PASS falls under a strict superset, whatever that cause turns out to be.

7.5 Generality and controls (summary)

Four follow-up experiments address three threat axes (source-pair generality, the context-length confound, and prompt-curability of the collapse); full protocols and numbers are in Appendix B. **(i) A second incomparable pair, different domain:** a private `cache`-module contract versus a key-normalization invariant is again verified incomparable on Haiku at $n = 40$ ($L_{\mathcal{D}} = +0.728 / +0.586$), over the same transport as the six-model record. The superset also collapses the contract cell, 35/40 to 0/40, at an upper bound of -0.62 once the budget is spread over the pair’s three contrasts, so the second domain carries a verified collapse of its own; on the trap cell the effect is mild and unverified, 29/40 to 26/40 (Section 8.4). **(ii) Length-matched padding:** an equally long filler arm stays at ceiling on both collapse cells (40/40, against W_{1+} ’s 0/40 and W_1 alone’s 39/40 and 31/40), which constrains the gross-length explanation on this model. **(iii) Routing instruction:** a one-line convention-routing note recovers most of the collapse ($0\% \rightarrow 82\%$ and $0\% \rightarrow 98\%$); writing that note requires knowing which source is decisive for which task family, which is exactly the coordinate structure the partial order formalizes, so the cure presupposes the diagnosis. **(iv) Structured context:** wrapping the same two conventions in labeled XML sections *without* any routing statement does not mitigate at all (0% on both cells, against 82% and 98% with the routing note), so segmentation alone does not cure the interference and the operative variable is routing knowledge. What the four arms show is a two-way contrast rather than a graded hierarchy: segmentation recovers nothing at all, while routing recovers most of the collapse and, on `api_argorder`, matches or exceeds what W_1 alone achieves in this control (39/40 against 31/40; the one-sided 90% lower bound is $+0.0095$, while a two-sided 90% interval contains zero). Read as a mechanism this points at convention-to-task routing rather than convention segmentation, but it is one model at $n = 40$ and the reading is an interpretation of the contrasts, not a verified claim.

7.6 The engineering consequence: four selection policies, one measured table

Table 5 scores four selection policies per model as mean PASS over the four verified cells of Table A1. *Best fixed source* grants the entire query-independent-scalar class its best case: any such scalar induces one total order, hence one top-1 reused for every task, and we let it pick the oracle-best source (W_1 for every model). *Topical top-1* re-scores per task with topical relevance, choosing W_1 on the API cells and W_2 on encoding, but also W_2 on the trap, where every measured topical ranker puts the wrong source first (see Section 7.4.2). *Concatenate both* refuses to choose and submits W_{1+} . *Oracle over $\{W_1, W_2\}$* takes the per-cell maximum over the non-dominated pair. It is scored on the same outcomes it maximizes over, so it is an oracle and not an achieved result: a deployable router would have to be fixed before testing and evaluated on held-out tasks under matched latency and token budgets, which this paper does not do. That column is selection *headroom* on this battery, not a delivered gain: the oracle’s 93.0% mean against 71.5% (best fixed source), 68.3% (topical top-1) and 47.9% (concatenation).

Three structural facts, not the magnitudes, are the point. First, the two scalar policies fail on *different* cells, the fixed source can never carry encoding, topical top-1 forfeits the trap, yet differ by only 3.1 points on average, and not at all on three of the six models: the cap is the policy *class*, since any total order must abandon at least one cell, not the choice of scorer (Theorem 2, Proposition 2). Second, the policy that refuses to choose is the *worst* of the four: anti-monotonicity is the dominant

Table 5: Mean PASS (%) over the four verified cells under four selection policies, per model, computed from the per-cell PASS counts behind Table A1 (averaging the rounded percentages instead would shift a column by up to 0.4). The last column is the per-cell maximum of W_1 and W_2 *read off the test outcomes*: an empirical oracle over the pair, not a router fixed in advance and then evaluated, and not a confidence bound on any policy’s population quality. It is also not a ceiling over all selection policies, since it excludes W_{1+} as a candidate: admitting it raises the Haiku oracle from 82.5 to 95.0, almost all of it from the encoding cell, where the pair’s best is 55% while W_{1+} reaches 100%, worth 11.25 points of the four-cell mean against 1.25 from the trap cell.

Model	Best fixed source	Topical top-1	Concatenate (W_{1+})	Oracle over $\{W_1, W_2\}$
Haiku-4.5	68.8	58.8	50.0	82.5
Sonnet-4.6	75.0	75.0	50.0	100.0
Opus-4.8	75.0	75.0	47.5	100.0
GPT-5.5	75.0	75.0	50.0	100.0
DeepSeek-V4-Pro	60.0	52.5	41.9	76.9
Kimi-K2.6	75.0	73.8	48.1	98.8
Mean	71.5	68.3	47.9	93.0

cost (see Section 7.4.3). Third, the oracle shortfall from 100% is neither pure selection error nor pure capability. For DeepSeek (68–75% band) it is residual failure under the prompts and decoder we tested, which is not a bound on what the model can do. For Haiku the largest remaining shortfall is on `enc_amount` (22/40), and on `api_argorder` (32/40) the retained completions (see Section 7.4.5) show the `import ledger` line the task needs missing from 7 of that arm’s 40 replies, a packaging effect rather than a selection error. The four cells were built to exhibit the phenomenon, so these magnitudes diagnose the failure structure rather than estimate production-traffic gains.

What follows for practice. Four statements survive the caveats above. (i) One number per source is structurally lossy: when two sources each carry a decisive fact, any single task-independent ranking drops one of them (Theorem 2). Keep both and route by task instead; the per-cell maximum in Table 5 is headroom read off the test outcomes, and no implementable router here captures it, on these tasks or new ones. Concatenating them is not forbidden by the theorem, but on this pair it is the worst of the four policies measured (Table 5). (ii) Do not cache a source’s *value* across task families; cache retrieval representations instead, because the value is model- and task-relative (see Section 7.4.4). (iii) Adding correct, on-topic text to a working prompt can break it, by as much as 100% $\rightarrow$ 0%, so a regression after enlarging the context is a candidate anti-monotonicity collapse before it is a model failure (see Section 7.4.3). (iv) A cheap topical ranker fails exactly where the surface lexicon points at the wrong source, which is where a listwise reader earns its cost (Proposition 2). Each is measurable on any stack with the released harness.

8 Discussion

The results recast a piece of practitioner folklore, “relevance does not transfer across tasks”, as a structural fact: when two sources are Blackwell-incomparable the order over them is partial, and any per-source scalar score, by imposing one task-independent preorder, must fail to rank them faithfully on some task. What follows concerns what that evidence does and does not license, and where the formalization is a model of the realized object rather than a proof about it.

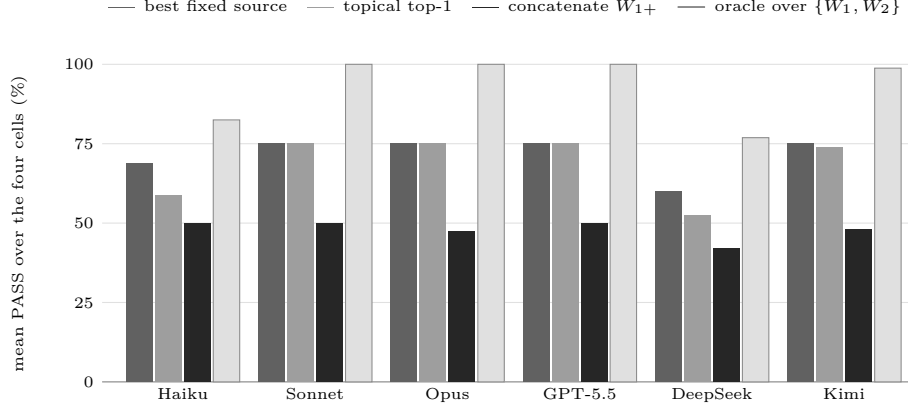


Figure 3: Mean PASS over the four verified cells under the four selection policies of Table 5, per model. Two things are visible that the table states in prose. The two scalar policies land within about a point of each other on most models while failing on *different* cells, so the ceiling is the policy class rather than the choice of scorer. And concatenating both sources is the *worst* of the four on every model, below even a fixed single source: refusing to choose costs more than choosing wrongly. The last column reads the per-cell maximum off the test outcomes and is an oracle over the pair, not a router fixed in advance and then evaluated.

8.1 Scope of the claims

Three limits of scope govern everything above. **(i) Two registers.** The theorems are about the ideal user of a source; every number we measure is the pass rate of one fixed, non-optimal rule δ_M . The two registers are established separately, not one from the other. Structural incomparability follows from the projection construction and Lemma 1, with no prior, no caps and no measurement. The experiments separately verify realized cross-advantage for the fixed model on the named task pairs (Assumption 1), which is $(M, \mathcal{D})$ -incomparability and nothing more. Lemma 2 is a conditional Bayes-value criterion whose guessing-cap hypotheses these state-conditional measurements do not estimate. So the measured crossover does not *earn* the structural conclusion, and need not: for these sources Lemma 1 already supplies it by construction. **(ii) The estimator’s name.** $\hat{\delta}_{\mathcal{D}}$ does no minimization over garblings and ranges over a fixed set of tasks; it is a one-sided *value-deficiency surrogate*, not a Le Cam deficiency, and the two are not interchangeable for a fixed rule (Remark 8). What is rigorous is the verification, which we regard as the most robust part and which stands whether or not the deficiency analogy holds. **(iii) What the verification covers.** It is correct because it pays for picking the worst task with exact intervals and a union bound; its support is small (two of the four cells carry each model’s result, and which two is model-specific: an API cell and the encoding cell on five models, the trap cell rather than an API cell on DeepSeek-V4-Pro), each family is controlled simultaneously at level η over the intervals it names, while the conjunction *across* families carries no single joint guarantee (see Section 6.5), and we call $\mathcal{D}$ a fixed adversarial battery, not a task distribution.

8.2 Anti-monotonicity is real, verified, but not capability-ordered

The harm object is the direct contrast $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$, which measures whether appending the (irrelevant-to-this-task) W_2 to a sufficient W_1 degrades performance (the second difference Ψ_T is an interaction term, not a harm measure; Remark 9). We verify $\Delta_T \leq -0.30$ on at least one cell for all six models (Table 4); nine of the twelve cells are a stark $40/40 \rightarrow 0/40$

(upper bound -0.85), on both API cells of Sonnet, Opus, GPT-5.5 and Kimi and on Haiku’s `api_post_ok`. This is a verified empirical falsification of *superset-monotonicity* for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone scalar context-value score can represent.

Three cautions attach to it. The harm is robust across the capability ladder but not ordered by it (see Section 7.4.3), so we claim no capability trend. Which cell carries the verification is still a choice: we fix it per model before unblinding rather than picking the starkest afterwards, and cells differ in their W_1 ceiling, so magnitudes are not comparable across models even though each binary verdict stands on its own. And the two Opus rows join arms from different collection windows (see Section 7.3); the harm contrasts do not, since each is read within a single run.

8.3 Mechanism: a correlate, not an explanation

On both collapse cells of all six models the harmful arm inflates output, between $+54\%$ and $+321\%$ (see Appendix C). On the still-passing encoding cell the same superset arm ranges from -73% to $+41\%$ with no consistent direction, so the association does not establish an internal cause and we treat it as a reproducible correlate of the harmful arm rather than a mechanism. Per-model figures are in Appendix C.

8.4 Limitations and threats to validity

We collect the scope restrictions here; each is stated once and sized to its weight.

From diagnosis to selection. The constructive selection rule the partial order implies, keep the verified non-dominated set and cover what it leaves uncovered, is left for future work; here we note only its cost. Building the verified order on m candidates costs $m \cdot |\mathcal{D}| \cdot n$ verifier-gated calls, curation scale, not corpus scale; scaling verified selection to retrieval over 10^6 passages, by amortizing the probe into a learned deficiency predictor or running the partial-order step as a re-ranking head over a scalar shortlist, is the central open problem this work opens. Keeping a non-dominated set instead of a top- k prefix already exists as an operator, the skyline retrieval step of Zhu et al. [7], and learned set selectors optimize evidence sets end to end [26, 27]; what the partial order adds is the object the set is taken over, verified task value rather than heuristic per-item scores. We do not claim that retaining the non-dominated set makes downstream selection *safe* under anti-monotonicity, and nothing here proves it: the verification bounds the relation *between* the candidates, not what a selector then does with them, and concatenating two retained members is the move our own record shows can lower PASS (see Section 7.4.3). What the partial order contributes at this level is diagnostic.

Source-pair generality. Incomparability and dominance hold for two hand-built pairs in two domains, both measured over the same transport as the six-model record. On the second pair the verification returns $L_{\mathcal{D}} = +0.728 / +0.586$, so the crossover is present in a `cache` module as well as a `ledger` one, and the superset carries a verified collapse there too: it takes the contract cell from 35/40 to 0/40, upper bound -0.62 with the budget spread over the pair’s three contrasts. The other two cells do not collapse; the trap cell moves only 29/40 to 26/40, nowhere near the $\tau = 0.30$ margin. An earlier command-line measurement of this pair behaved differently enough that we reported it as regime-dependent; that caveat concerned a transport the record no longer uses. Two gaps remain. The cross-vendor record is still single-pair: five of the six models were never run on it. And the pair did not reproduce the relevance-trap effect at all, its prompt not being lexically

V_2 -leaning, so the strong mis-rank is still a one-pair result. A third domain and a vendor-general strong mis-rank remain open next steps.

Cross-vendor magnitudes are still only suggestive. All six models are now run the same way, one HTTP request per draw at $n = 40$ (see Section 7.3), so neither the turn budget nor the transport confounds the comparison. What remains is the ordinary caution that different vendors’ models differ in ways we do not control (training data, size, tokenizer), so the *magnitude* comparisons (the value spectrum, the Δ_T ladder) should be read as suggestive; the binary within-model claims do not depend on any cross-vendor comparison.

The construction is an idealization. It models sources as clean projections onto independent coordinates; real chunks overlap, contradict, and leak (the measured θ -dependence is exactly such leakage), so it is the cleanest *sufficient* condition for incomparability, realized exactly by our hand-built pairs and not claimed of arbitrary ones.

Prevalence is not measured, and is the decisive next experiment. Every claim here is existential: it establishes *that* verified incomparability and superset harm occur, on which models, and with which confounds excluded. It does not establish *how often* either occurs in a deployed retrieval stream, and no number in this paper should be read as an estimate of that. The four cells were built to make the structure visible, so their aggregate magnitudes are diagnostic of the failure mode rather than of production traffic. The direct experiment is specified: draw document pairs from a natural corpus, keep the procedure of Section 6 but write tasks and verifiers that corpus admits, since the four ledger and wire tests do not transfer, and report what fraction verifies incomparable and harmful, with the sampling frame stated. That fraction converts into a lower bound on prevalence without a union bound over pairs. Write π for the probability that a drawn pair has the property and V for the event that the procedure verifies it. The procedure is one-sided at level η , so $\Pr[V \mid \text{not } I] \leq \eta$ and therefore $\Pr[V] \leq \pi + (1 - \pi)\eta$, which rearranges to $\pi \geq (\Pr[V] - \eta)/(1 - \eta)$. Replacing $\Pr[V]$ by an exact Clopper–Pearson lower bound $\underline{p}$ on the verified fraction over the sampled pairs gives $\pi \geq \max\{0, (\underline{p} - \eta)/(1 - \eta)\}$ at confidence $1 - \alpha$: the correction applies to one marginal probability, not once per pair, because nothing here asserts that each individual verified pair is a true positive. Three conditions bind. The pairs and the procedure’s outcomes must be i.i.d. draws from the stated frame; adaptive selection breaks that, and sharing documents across pairs may introduce dependence. The property I is whatever the certificate actually verifies—PASS-incomparability for a named model on a named task battery, not unconditional Blackwell incomparability of the underlying sources. And a joint claim, “incomparable *and* harmful,” needs I , V and the one-sided error η redefined for that conjunction rather than reusing these. We name prevalence as the open question rather than approximate it here, because a figure computed from hand-built pairs would be circular.

Scope of the behavioural controls. The padding, routing-note and structured-context arms, and the reversed-order arm, were each run over the same API path as the main record, on Haiku at $n = 40$ on the two collapse cells, in the same regime and on the same task versions as the main table. None of the four is part of the six-model record, so what these controls establish is established on one model, not across the roster. The padding match is by *characters* (1108 against 1095 at the arm level), not by subword tokens, and because W_2 is symbol-dense while the filler is plain prose the two arms are not tokenizer-matched; token count is therefore *not* isolated, only gross length. The filler is also off-topic for the `ledger` module. Taken together, on this model the added block’s content

and the routing instruction move the outcome, while equal character-length off-topic filler does not. The controls constrain positional and gross-length explanations in the tested configuration rather than eliminating them: they do not isolate tokenization or processing difficulty, do not identify a unique cause, and do not exclude every position-dependent mechanism. Reversing the two blocks leaves both cells at 0/40, so on this record order changes neither the sign nor the size of the harm, which constrains a positional account without ruling one out for configurations we did not test. The mechanism and refusal counts of Appendix C, with the non-empty extraction record, are likewise automated flags over retained draws: they document what the failing replies contain, establish no universal causal mechanism, and do not exclude every packaging problem. A tokenizer-matched control and an on-topic but verifier-irrelevant control remain untested, as does any of these across the roster. The order control has been run over the same API path as the rest of the controls, but on one model and on the two collapse cells only (see Section 7.4.3).

Sampling and verification regime. The verification assumes n i.i.d. Bernoulli PASS draws per cell (Assumption 2). Each draw is a separate call with no shared session or conversation history (protocol in Appendix D), which makes independence plausible; it does not *establish* it, nor that the served backend was stationary across a run. Both remain assumptions on which the coverage statements depend, and a provider-side model or routing change during a run would break them silently. Stationarity *between* runs rests on less still, and the cross-window contrasts of Section 7.3 depend on it; the within-run harm contrasts do not. Marginal-versus-joint control is discussed in Section 8.1.

Ranker spectrum and synergy. The trap mis-rank spans lexical cosine and three representative open-weight learned rankers, with an LLM listwise reranker ranking the pair correctly, which the theory permits rather than predicts, but the learned checkpoints are the small members of their families and the evidence is one trap in one domain. The mis-rank is not evidence of class membership: we do not verify that any learned checkpoint’s score is invariant to the verifier-decisive coordinate or increasing in lexical overlap alone, so Proposition 2 is not shown to bind these rankers. Positive interference (synergy) is untested by construction: W_1, W_2 are task-disjoint, so a complementary pair would be needed to test whether a superset can also *help* beyond its parts; we make no claim about it.

9 Conclusion

Context selection is a partial-order problem, not a scalar one. Blackwell’s comparison of experiments identifies a structural limitation of task-independent context rankings: when two sources are incomparable, no one per-source score fixed across tasks ranks them correctly for every decision problem (Theorem 2). For *fixed* model rules we measure a separate, task-restricted PASS relation with finite-sample bounds, tied to the classical order only through the conditional bridge of Lemma 2. On six models across four vendors both directions of PASS-incomparability stay positive under a joint 90% guarantee; on each of them a deployed relevance score ranks the pair the wrong way on a designed trap; and adding a strictly larger source lowers PASS on all twelve tested API contrasts, eleven of which meet the 0.30 harm margin under joint 90% control, with every model retaining at least one verified contrast. What a private source is worth is model-relative: guessability runs from 0% to 100% across the six models, one of which solves the task with no context at all, and what it tracks these runs cannot identify. Bringing Blackwell and Le Cam to inference-time context is not ours to claim; concurrent work reached the same apparatus from the admission side (see

Section 2.3). What follows from taking *incomparability* rather than deficiency-minimization as the object is the impossibility, and the instrument that tests it.

The scope is narrow and stated exactly. Theorem 2 and Lemma 2 concern the idealized Bayes-optimal user, while Theorem 3 concerns the fixed rule a model realizes and invokes no such user at all; the measured crossover and collapse are properties of those *fixed* rules; the two sources' incomparability is structural, resting on the projection construction and Lemma 1 rather than on measurement; Lemma 2 is a conditional Bayes-value criterion whose guessing caps we do not estimate; and the estimator is a regret-gap surrogate, not a Le Cam deficiency. All six models are measured under one protocol, so cross-vendor magnitudes carry no transport confound; they remain read with caution as model differences and collection windows (see Section 7.3); the verified claims are within-model contrasts. What the impossibility excludes is one number per source fixed across tasks and query-conditioned scores meeting the two hypotheses of Proposition 2; a task-conditioned score, a router over public metadata, or a listwise reader that pays to read the decisive content lie outside it (Remark 6). The behavioural controls constrain block-order and gross-length explanations on one model; they do not establish an internal mechanism. The decisive next test is prevalence: draw source pairs from a natural corpus rather than building them, run the same verification unchanged, and report how often a deployed retrieval stream produces a pair that verifies incomparable (see Section 8.4).

Author Contributions

Conceptualization, A.B. and V.L.; methodology, A.B., V.L. and P.P.; software, A.B.; validation, A.B., V.L. and Z.R.; formal analysis, A.B. and P.P.; investigation, A.B.; resources, V.L.; data curation, A.B.; writing—original draft preparation, A.B.; writing—review and editing, A.B., V.L., P.P. and Z.R.; visualization, A.B. and Z.R.; supervision, V.L. and P.P.; project administration, V.L. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no external funding. The measurement costs reported in Appendix D were borne by the authors.

Institutional Review Board Statement

Not applicable. The study involved no human participants and no animals; all measurements are automated queries to commercial language-model endpoints, graded by executable verifiers.

Informed Consent Statement

Not applicable.

Data Availability Statement

All data and code supporting the reported results are openly available at <https://github.com/abilous-ti/blackwell-llm-context>. The repository contains the measurement harness (Python

standard library only), every per-model run log and result file cited in Appendix B, and the raw-completion diagnostic records of Section 7.4.5. Two kinds of record must be distinguished. The original runs stored *aggregated* per-cell counts and logs; raw model completions were not retained, and cannot be recovered. The retained completions of Appendix C.1 and the diagnostic runs stored *raw* completions, and those are released in full. Claims that depend on reading replies are therefore made only against the second kind. **results/MANIFEST.md** pins every released record by SHA-256 and maps each table and figure to the files it is computed from; it certifies the integrity of what is published, not the existence of what was never kept. The repository’s full commit history is published with it, so the provenance of every artifact is inspectable. No data were withheld; every reported figure is computed from those files by the released scripts.

Acknowledgments

Anthropic’s Claude models were used for code scaffolding, experimental-plan review, implementation assistance, consistency and artifact audits, manuscript organization, and L^AT_EX formatting. OpenAI’s Codex was used separately to audit the manuscript against the released records; several errors it found, including a diagnostic rule that under-reported what it was testing for, are corrected in this version, and each correction was checked by re-running the released scripts. The work spans many sessions and the assistant version in use was not logged throughout, so we identify the tools by family rather than assert version strings we did not record. The authors inspected code changes, executed the experiments, verified outputs against frozen manifests and source files, checked cited sources, and take full responsibility for the study design, results, interpretations, and manuscript.

Use of Generative Artificial Intelligence

Language models appear in this work in two distinct roles, and we separate them because only one bears on the results. *As the object of study*: the measured quantities are pass rates of commercial language models, obtained through the interfaces described in Section 7.3; everything else we report, the confidence bounds, the ranker scores, the token and cost figures, is computed from those runs. *As a tool*: generative-AI tools assisted with the tasks disclosed in the Acknowledgments, analysis and interpretation among them. Every numerical result was computed by the released scripts from the released records, which **results/MANIFEST.md** pins by SHA-256 and maps to the table or figure that uses it, and was checked by the authors, who retain responsibility for the analysis and the conclusions. The Data Availability Statement says which records are aggregate-only and which retain raw completions.

Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BSS	Blackwell–Sherman–Stein (theorem)
CP	Clopper–Pearson (exact binomial confidence interval)
DPP	Determinantal point process
EIG	Expected information gain
IR	Information retrieval
LLM	Large language model
MMR	Maximal marginal relevance
PASS	Pass rate of a fixed model rule on a verifier-graded task
PRP	Probability ranking principle
RAG	Retrieval-augmented generation

A Deferred proofs and technical material

Proof of Lemma 1

Proof. (a) For deterministic experiments a garbling G with $\kappa_{W_B} = G \circ \kappa_{W_A}$ must compute $\pi_B(S)$ from $\pi_A(S)$ alone (no further access to S). Such a measurable map exists iff π_B is $\sigma(\pi_A)$ -measurable, i.e. $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$. Under coordinate independence and non-degeneracy, $\sigma(\pi_B) \subseteq \sigma(\pi_A)$ holds iff every coordinate in B is among those in A , i.e. $B \subseteq A$: if some $j \in B \setminus A$, then S_j is independent of $\pi_A(S)$ and non-constant, so it is not $\sigma(\pi_A)$ -measurable, and no garbling can reconstruct it; conversely if $B \subseteq A$ the coordinate-deletion map is a (deterministic) garbling. (b) is the contrapositive of (a) applied in both directions. $\square$

Proof of Lemma 2

Proof. (a) Suppose, for contradiction, $W_2 \succeq_B W_1$. By Theorem 1, $V^*(W_2, T) \geq V^*(W_1, T)$ for every task T , in particular at T_a . By the sufficiency hypothesis, T_a 's verifier is determined by its decisive coordinate, which lies in $A_1 \setminus A_2$ and is revealed exactly by W_1 (Definition 3). Solvability, the hypothesis's separate half, supplies μ_{T_a} -almost surely an action the verifier accepts, measurable in that coordinate and the facts known μ_{T_a} -almost surely. Reading the coordinate off W_1 's signal and playing that action attains $v_{T_a} = 1$ μ_{T_a} -almost surely, so $V^*(W_1, T_a) = 1$. By hypothesis $g_a = V^*(W_2, T_a) < 1$, hence $V^*(W_1, T_a) > V^*(W_2, T_a)$, a contradiction. Therefore $W_2 \not\succeq_B W_1$; symmetrically $g_b < 1$ forbids $W_1 \succeq_B W_2$, so W_1, W_2 are Blackwell-incomparable.

Two things this argument does *not* use are worth stating, because the natural shorter proof uses both and is wrong. It never asserts $V^*(W_1, T_a) \geq \text{PASS}_M(W_1, T_a)$: the left side is a Bayes value under μ_{T_a} , the right a pass rate at the single realized convention, and optimality relates V^* only to the μ_{T_a} -averaged $\overline{\text{PASS}}_M$ (Remark 1). And it never treats the measured crossover as the hypothesis or as an estimate of it: the caps are assumed, and nothing reported here determines g_a or g_b . In the converse direction the caps keep apart two failures a bare crossover cannot distinguish: opposed PASS_M rates on a strictly *comparable* pair, which is what (W_1, W_{1+}) gives, with $W_{1+} \succeq_B W_1$ strictly and Theorem 3's non-coherence a monotonicity violation; and opposed rates under Blackwell *equivalence*, illustrated without a certified instance by the padding arm (W_1, W_{pad}) (Remarks 3 and 2). Neither licenses a structural conclusion from a crossover alone.

(b) Assumption 1 directly yields $\text{PASS}_M(W_1, T_a) > \text{PASS}_M(W_2, T_a)$ and the reverse at T_b , which is Definition 5's incomparability. The unconditional converse fails, witnessed by the decoder that ignores its context: whatever the sources, $\text{PASS}_M(W_1, T) = \text{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$, so $W_1 \succeq_{M, \mathcal{D}} W_2$ and $W_2 \succeq_{M, \mathcal{D}} W_1$ both hold and the pair is $(M, \mathcal{D})$ -comparable however Blackwell-incomparable it is. A monotonicity violation on a *comparable* pair is a different phenomenon, not by itself such a witness. $\square$

Proof of Theorem 2

Proof. By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, “not $W_1 \succeq_B W_2$ ” yields a task T_b (a prior plus bounded utility) with $V^*(W_2, T_b) > V^*(W_1, T_b)$, and “not $W_2 \succeq_B W_1$ ” yields a task T_a with $V^*(W_1, T_a) > V^*(W_2, T_a)$; both inequalities are *strict*. These witnesses carry bounded utility and need not be binary verifiers (Definition 2); for projection sources they may be taken binary (proof of Lemma 2(a)). A query-independent φ assigns the single ordered pair $(\varphi(W_1), \varphi(W_2))$, fixing one task-independent total *preorder* on $\{W_1, W_2\}$. If $\varphi(W_1) = \varphi(W_2)$ it expresses no strict preference and is unfaithful on both; otherwise the strict order it fixes agrees with the optimal-value order on at most one of T_a, T_b (they point opposite ways) and is violated on the

other. Either way faithfulness fails on at least one, which is the claim, and a fixed task-independent tie-breaking rule only removes the first case. The PASS_M conclusion follows by replacing V^* with PASS_M via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin γ . $\square$

Proof of Proposition 2

Proof. For (a), $\varphi = g \circ \psi$ and $\psi(W_1, T_c) = \psi(W_2, T_c)$ force equal scores, while faithfulness on T_c would require a strict gap. For (b), $\text{ov}(W_2, T_c) > \text{ov}(W_1, T_c)$ with g strictly increasing gives $\varphi(W_2, T_c) > \varphi(W_1, T_c)$, while utility favours W_1 because T_c 's PASS requires W_1 's coordinate; the two orders disagree. Part (b) makes explicit the two things the informal reading of “saturated lexicon” presumes, strict monotonicity and overlap being the only feature read, without which no strict inequality follows. $\square$

Proof of Proposition 1

Proof. Let G be the garbling with $\kappa_{W_2} = G \circ \kappa_{W_1}$. Any predictor $g \in \mathcal{V}$ acting on W_2 's signal is realized on W_1 's signal by the composite $g \circ G$, which lies in $\mathcal{V}$ by the closure hypothesis. Therefore the best $\mathcal{V}$ -predictor under W_1 attains at least the predictive accuracy of the best under W_2 , so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under W_1 . The scalarity is immediate: $I_{\mathcal{V}}$ returns a single real number per source, so it induces one task-independent total preorder and, by Theorem 2, cannot rank a Blackwell-incomparable pair faithfully on all tasks. $\square$

Proof of Proposition 3

Proof. Let E be the event that all $2|\mathcal{D}|$ Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most $\alpha = \eta/(2|\mathcal{D}|)$ (exact coverage of Clopper–Pearson [9]), so by the union bound $\Pr[E^c] \leq 2|\mathcal{D}| \cdot \alpha = \eta$, giving $\Pr[E] \geq 1 - \eta$. On E , for every T we have $p_{W_2, T} \geq L_{W_2, T}$ and $p_{W_1, T} \leq U_{W_1, T}$, so $p_{W_2, T} - p_{W_1, T} \geq L_{W_2, T} - U_{W_1, T}$; taking $\max_T$ on both sides, $\sup_T [p_{W_2, T} - p_{W_1, T}] \geq L_{\mathcal{D}}(W_1, W_2)$. Thus on E (probability $\geq 1 - \eta$) the true sup is at least $L_{\mathcal{D}}(W_1, W_2)$, which is the claimed lower confidence bound; if it is positive, the true sup is positive, verifying $d_{M, \mathcal{D}}(W_1, W_2) > 0$. The reverse direction uses $L_{W_1, T}$ and $U_{W_2, T}$, which are endpoints of the *same* $2|\mathcal{D}|$ intervals already included in E ; therefore both one-sided verifications are valid on the single event E , and their conjunction holds at $\geq 1 - \eta$ without a further Bonferroni factor. $\square$

Proof of Proposition 4

Proof. Only two quantities enter $\bar{\Delta}_{T^*}$: an upper endpoint on p_{W_{1+}, T^*} and a lower endpoint on p_{W_1, T^*} . The two arms are independent Bernoulli samples (separate completions). Let E_+ be the event $p_{W_{1+}, T^*} \leq U_{W_{1+}, T^*}$ and E_- the event $p_{W_1, T^*} \geq L_{W_1, T^*}$. By the exact one-sided coverage of Clopper–Pearson at noncoverage $\alpha = \eta/2$ per endpoint, $\Pr[E_+^c] \leq \eta/2$ and $\Pr[E_-^c] \leq \eta/2$, so by the union bound $\Pr[E_+ \cap E_-] \geq 1 - \eta$. On that event

$$\Delta_{T^*} = p_{W_{1+}, T^*} - p_{W_1, T^*} \leq U_{W_{1+}, T^*} - L_{W_1, T^*} = \bar{\Delta}_{T^*},$$

which is the claim; if $\bar{\Delta}_{T^*} \leq -\tau$ then $\Delta_{T^*} \leq -\tau$ at confidence $\geq 1 - \eta$. Note that the budget is spent on exactly the two endpoints used: a two-sided interval at level $\eta/2$ would place $\eta/4$ in each tail, leave two endpoints unused, and deliver $1 - \eta/2$ while widening the bound, which is why the

allocation is stated per endpoint rather than per interval. For a post-hoc selected cell, replacing α by $\eta/(2|\mathcal{D}|)$ and union-bounding over the $2|\mathcal{D}|$ endpoints preserves $\geq 1 - \eta$ simultaneous coverage. $\square$

Proof of Theorem 3

Proof. $W_{1+} \succeq_B W_1$ by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)); write G for that deletion, so $G \circ \kappa_{W_{1+}}(\cdot \mid s) = \kappa_{W_1}(\cdot \mid s)$ for every state s . The rule $\delta_M \circ G$ is then Blackwell-coherent on T^* in the sense of the theorem, and its realized pass rate on W_{1+} equals $\text{PASS}_M(W_1, T^*)$ at the realized state, since the two rules see the same signal law. The hypothesis $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$ therefore says that δ_M on W_{1+} is strictly beaten by its own pruned version on the same input, so δ_M is not Blackwell-coherent on T^* . Note what this argument does *not* use: it is the simulation construction behind (i) $\Rightarrow$ (ii) of Theorem 1 applied state-conditionally, not an instance of statement (ii), which is Bayes-averaged and so says nothing about a single state (Remark 1). A monotone scalar ϕ assigns $\phi(W_{1+}) \geq \phi(W_1)$ (monotonicity), hence orders W_{1+} above W_1 and cannot reproduce the realized reversal. $\square$

Relation to LLM-evaluation statistics

Remark 10 (Relation to LLM-evaluation statistics). The verification sits in a small but growing statistical-methodology literature for black-box model evaluation: paired-difference designs and clustered standard errors for evals [70], two-sample tests verifying behavioral differences between API-served models [71], and the recommendation to prefer exact frequentist intervals over CLT approximations when per-cell n is below a few hundred [72], which is precisely our regime ($n \in [20, 80]$ per cell) and precisely what the exact Clopper–Pearson construction provides.

Power and the role of n

Both verifications are driven by how close the arms sit to an endpoint. For a stark cell with empirical proportion at an endpoint ($\hat{p} \in \{0, 1\}$) the Clopper–Pearson interval becomes one-sided and clears 0 even at small n . It does not become a point: at $n = 40$ and two-sided level $\eta/8$, $0/40$ gives $[0, 0.119]$ and $40/40$ gives $[0.881, 1]$, and it is that 0.119 the crossing has to beat. The incomparability verification (6) is carried by exactly such cells (W_1 at $\approx 100\%$, W_2 at 0% and vice versa). The harm contrast (8) needs the gap between two arms to exceed τ after both intervals are paid for, and a negative point estimate need not verify either the sign of the harm or the margin. On the second source pair’s trap cell PASS moves from $29/40$ to $26/40$, but the one-sided 90% upper bound on the difference is $+0.188345$: neither $\Delta_T < 0$ nor $\Delta_T \leq -0.30$ is verified by that contrast.

B Extended experimental record

Generality and controls: full protocols

Four follow-up experiments address three threat axes: source-pair generality, the context-length confound, and whether the anti-monotonicity collapse is curable by cheap prompting.

How the bounds below are to be read. A bound on a two-arm difference is assembled from the two proportions’ exact intervals. Each endpoint quoted here is a *separate one-sided* 90% bound, which is the form Proposition 4 uses and the form in which each harm claim is made; the two endpoints together carry a guaranteed joint coverage of at least 80% under this union-bound argument, so where a two-sided statement is wanted we assemble it from individual 95% Clopper–Pearson regions

Table A1: Per-cell PASS rates (%) for all four verified cells, by arm, all one HTTP turn at $n = 40$. Every model was measured on every cell and arm. The encoding rows (`enc_amount`) carry the $W_2 > W_1$ direction of incomparability: on every model W_2 solves encoding while W_1 fails it, mirroring the API cells where W_1 solves and W_2 fails.

Model	Cell	none	W_1	W_2	W_{1+}
Haiku-4.5	<code>api_post_ok</code>	0	100	0	0
Haiku-4.5	<code>api_argorder</code>	0	80	0	0
Haiku-4.5	<code>enc_amount</code>	0	0	55	100
Haiku-4.5	<code>trap_store_wire</code>	0	95	0	100
Sonnet-4.6	<code>api_post_ok</code>	0	100	0	0
Sonnet-4.6	<code>api_argorder</code>	0	100	0	0
Sonnet-4.6	<code>enc_amount</code>	0	0	100	100
Sonnet-4.6	<code>trap_store_wire</code>	0	100	0	100
Opus-4.8	<code>api_post_ok</code>	0	100	0	0
Opus-4.8	<code>api_argorder</code>	100	100	0	0
Opus-4.8	<code>enc_amount</code>	0	0	100	90
Opus-4.8	<code>trap_store_wire</code>	0	100	0	100
GPT-5.5	<code>api_post_ok</code>	5	100	0	0
GPT-5.5	<code>api_argorder</code>	75	100	0	0
GPT-5.5	<code>enc_amount</code>	55	0	100	100
GPT-5.5	<code>trap_store_wire</code>	0	100	0	100
DeepSeek-V4-Pro	<code>api_post_ok</code>	0	75	0	0
DeepSeek-V4-Pro	<code>api_argorder</code>	2	68	0	5
DeepSeek-V4-Pro	<code>enc_amount</code>	0	0	68	65
DeepSeek-V4-Pro	<code>trap_store_wire</code>	0	98	0	98
Kimi-K2.6	<code>api_post_ok</code>	0	100	0	0
Kimi-K2.6	<code>api_argorder</code>	45	100	0	0
Kimi-K2.6	<code>enc_amount</code>	0	0	95	92
Kimi-K2.6	<code>trap_store_wire</code>	0	100	0	100

and say so. Decisions use unrounded endpoints; displayed confidence bounds are rounded outward, lower bounds downward and upper bounds upward.

A second incomparable pair, different domain. We repeated the protocol with a second hand-built pair on a private `cache` module: V_1 , the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False “newly inserted” flag, a `CacheError` type), versus V_2 , a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at $n = 40$ the pair is again verified incomparable ($L_D = +0.728/ +0.586$, both directions clear of zero) with a clean double crossover (V_1 solves the contract and trap cells, V_2 solves key-normalization, all none-baselines floor at 0/40). Anti-monotonicity reproduces as a verified collapse on the contract cell: the superset takes `cache_put_ok` from 35/40 to 0/40, $\Delta_T = -0.87$ with an upper bound of -0.62 once the budget is spread over the pair’s three contrasts, clearing the margin $\tau = 0.30$ fixed for the whole programme. The other two cells do not: the superset lifts key-normalization from 0/40 to 40/40, and on the trap cell it moves 29/40 to 26/40, $\Delta_T = -0.075$, one-sided 90% bounds -0.3295 and $+0.1884$. The upper bound is positive, so this contrast verifies neither the sign of the harm nor the margin. The dominance control on the same run returns

$U_{\mathcal{D}}(W_{1+}, W_2) = 0.113$, inside the 0.15 tolerance, with a strict win the other way (0.65), so it covers the three-cell battery rather than a single cell. The relevance trap for this pair was a null, its prompt vocabulary leaning toward V_1 ($\varphi_{\text{lex}}(V_1) = 0.39 > \varphi_{\text{lex}}(V_2) = 0.25$ on the trap prompt), so relevance *agreed* with PASS and the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability, dominance and collapse results to a new domain*, not a second mis-rank.

A character-matched padding control. Anti-monotonicity compares W_1 against the strictly longer W_{1+} , and added input length alone is known to degrade performance [43], so the collapse could in principle be a token-count effect rather than a signal effect. To isolate this we added a padding arm W_{pad} : filler text in the same “team context (private)” register as W_2 , placed in the same position W_2 occupies in W_{1+} (padding first, W_1 last), character-matched to W_{1+} within 13 characters, but carrying no coordinate any verifier checks (office-logistics content). On Haiku at $n = 40$ on the two collapse cells, padding is free while the equal-length dominated signal is catastrophic: on `api_argorder`, W_1 31/40, W_{pad} 40/40, W_{1+} 0/40; on `api_post_ok`, W_1 39/40, W_{pad} 40/40, W_{1+} 0/40. The padded arm is never verified below W_1 alone: on `api_argorder` $\Delta_T = +0.225$ with a two-sided 90% interval of $[+0.0202, +0.3846]$, clear of zero, and on `api_post_ok` $\Delta_T = +0.025$ with $[-0.0875, +0.1316]$, consistent with no difference. Against the superset the same-run harm verification clears on both cells ($\Delta_T = -0.775$, one-sided 90% upper bound -0.5680 ; and -0.975 , upper bound -0.8146). In the tested configuration the collapse therefore does not follow from added length alone. The match is by characters, not subword tokens, and the filler is plain prose while W_2 is symbol-dense, so gross length is constrained but token count and processing difficulty are not. One residual axis is not isolated by this control: the padding is off-topic for the `ledger` module while W_2 is on-topic, so “topical but verifier-irrelevant” text remains a smaller, untested middle case.

A routing-instruction ablation: the cure presupposes the diagnosis. If the collapse reflects convention confusion, an explicit routing note might prevent it, and if a one-line prompt fully cured the harm, its practical significance would shrink. We measured this directly: the arm W_{1+}^{instr} appends to W_{1+} a single routing sentence stating which convention serves which task family (“for tasks that call the ledger API use only the API call contract; the wire-encoding invariant applies only to encode/decode tasks”). On Haiku at $n = 40$ the instruction recovers most, but not all, of the collapse: on `api_argorder`, $0/40 \rightarrow 39/40$, and on `api_post_ok`, $0/40 \rightarrow 33/40$; both recoveries over W_{1+} are large, with non-overlapping exact 95% intervals on the two proportions. Against W_1 alone the picture is cell-dependent and much weaker. On `api_argorder` the instructed superset scores 39/40 against 31/40, $\Delta_T = +0.200$, with a one-sided 90% lower bound of $+0.0095$; that separate one-sided statement clears zero, but the two-sided 90% interval assembled from individual 95% regions is $[-0.0232, +0.3839]$ and does not. On `api_post_ok` it scores 33/40 against 39/40, $\Delta_T = -0.150$, one-sided bounds -0.3025 and $+0.0281$, so neither superiority nor inferiority to W_1 is verified. The record shows that the instruction removes the collapse; it does not show whether the instructed superset matches W_1 alone. We pre-commit to the two readings jointly. First, the harm is substantially mediated by convention interference that explicit routing knowledge prevents, so pipelines that can attach per-task routing labels have a cheap mitigation. Second, and decisively for the paper’s thesis: *writing the routing sentence requires already knowing which source is decisive for which task family*, which is precisely the coordinate structure the partial order formalizes and the deficiency verification measures; the cure is an injection of selection-time coordinate knowledge into the prompt, not a scalar score, and no query-independent scalar can supply it (Theorem 2). Formally,

δ_M with the routing note is a *different, more Blackwell-coherent rule*; the verified falsification of Theorem 3 concerns the base rule and is unaffected. What the ablation does not settle is the selection-level question under the instructed rule: the instructed superset sits below W_1 alone on one cell and above it on the other, so on this record keeping the dominated signal is neither uniformly worse nor uniformly better than dropping it once routing knowledge is supplied. What the record does support is narrower: without that knowledge, forcing the superset is verified harmful, and supplying it requires the coordinate structure the partial order formalizes.

A structured-context ablation: labeled XML segmentation does not mitigate. To separate “the model cannot segment the two conventions” from “the model cannot route them to task families,” the arm W_{1+}^{xml} wraps the same two conventions, in the same order as W_{1+} , in labeled XML sections (`<wire_encoding_invariant>`, `<api_call_contract>`) with no routing statement. On Haiku at $n = 40$ per cell, structure alone does not mitigate at all: W_{1+}^{xml} scores 0/40 on both collapse cells, exactly the raw superset’s result, so labelled segmentation without a routing statement is not separated from the superset anywhere (both 95% intervals $[0.000, 0.088]$). The routing note is separated from both on both cells: 33/40 on `api_post_ok` ($[0.672, 0.927]$) and 39/40 on `api_argorder` ($[0.868, 0.999]$). On `api_argorder` the routing arm also exceeds this control’s own W_1 baseline of 31/40 ($[0.615, 0.892]$), so the ordering is not a graded ladder from superset to dropping the dominated signal; what the arms separate is segmentation, which changes nothing, from routing knowledge, which changes almost everything. As in Section 8.3 this localization is an interpretation of the behavioural contrasts, not a verified claim, and it rests on one model at $n = 40$, with the structure arm and the baselines it is compared against measured in one run.

C Experimental diagnostics

This appendix carries the per-draw diagnostic record behind Section 7.4.5 and the token observation of Section 8.3. Nothing here is required to read the verified results; it is the evidence for what the failing replies contain.

C.1 What the retained completions contain

Raw completions were kept for the three Anthropic models’ rows of the interference table; the other three models’ rows are aggregate-only. This appendix reads what was kept. The table below covers the two collapse cells of those three models, 40 completions per arm and cell, 480 in total; the same retention is in place for the behavioural controls and the second source pair.

Three screens follow and one thing is withheld. *The packaging screens*: no draw produced an empty extraction, and the `import` line is present in 39 or 40 of 40 replies everywhere except Haiku’s W_1 arm on `api_argorder` (33/40), which is the arm that passes, so those omissions depress the comparison arm and understate the harm rather than producing it. What this bears on is an empty extraction and a missing import, not every packaging explanation. *The harmful arm inflates output on every model*: 105 $\rightarrow$ 477 and 459 $\rightarrow$ 633 characters on Haiku, 94 $\rightarrow$ 383 and 180 $\rightarrow$ 253 on Sonnet, 93 $\rightarrow$ 286 and 100 $\rightarrow$ 252 on Opus. *The refusal screen comes back negative*: the phrase list below matches none of the 480 draws, so the refusal behaviour an earlier command-line record showed for Sonnet is not present here. Being a phrase list, it bounds explicit refusal language rather than every way a reply could decline.

The wire and refusal columns are two regular expressions, the first applied to the extracted code and the second to the prose outside the code fences:

Table A2: Retained completions on the two collapse cells, $n = 40$ per arm. “Import” counts replies carrying the `import ledger` line the task needs; “wire” counts replies matching the encoding rule below; “empty” counts replies from which no code block could be extracted; “refusal” counts replies declining the task. Mean length is in characters of the extracted code. Every column is reproduced by `harness/diag/retained_text_checks.py`.

Model	Cell	Arm	Mean length	Import	Wire	Empty / Refusal
Haiku-4.5	<code>api_post_ok</code>	W_1	105	40/40	0/40	0 / 0
		W_{1+}	477	39/40	40/40	0 / 0
	<code>api_argorder</code>	W_1	459	33/40	0/40	0 / 0
		W_{1+}	633	40/40	40/40	0 / 0
Sonnet-4.6	<code>api_post_ok</code>	W_1	94	40/40	0/40	0 / 0
		W_{1+}	383	40/40	40/40	0 / 0
	<code>api_argorder</code>	W_1	180	40/40	0/40	0 / 0
		W_{1+}	253	40/40	40/40	0 / 0
Opus-4.8	<code>api_post_ok</code>	W_1	93	40/40	0/40	0 / 0
		W_{1+}	286	40/40	40/40	0 / 0
	<code>api_argorder</code>	W_1	100	40/40	0/40	0 / 0
		W_{1+}	252	40/40	40/40	0 / 0

```
wire      re.search(r"#\{f\"{sign}\}|check\s*=% 10", code)
refusal   re.search(r"\bI can(?:no|')t\b|\bI won't\b|\bI'm (?:not able|unable)\b"
                  r"|\bunable to (?:help|assist|comply)\b|\bI must decline\b",
                  prose, re.I)
```

It matches the reply constructing W_2 ’s check-digit amount, whether as a literal or through an f-string, and it is quoted here because an earlier version of the same check looked only for a quoted literal and therefore reported zero on replies that plainly carry the encoding.

What is withheld is the mechanism. The flag separates the arms completely on the two API cells, 240/240 against 0/240 over the three models. It is not a correctness test: on those same cells it fires on 238/240 of the W_2 replies, every one of which fails the API verifier, and on 31/240 of the no-context replies. On `enc_amount`—a different population, in which applying the encoding *is* the contract—it fires on 109/120 of the W_2 replies while 102/120 of them pass. What the flag reports is an output pattern tied to the source condition; it does not identify an internal cause and says nothing about what the model represented internally. Theorem 3 does not depend on the answer: it constrains any rule whose PASS falls under a strict superset, whatever the internal cause.

C.2 Output-token inflation on the harmful arm

Every collapse is accompanied by output-token inflation on the harmful arm. Across the twelve API collapse cells W_{1+} emits between +54% and +321% more output tokens than W_1 (Haiku +321%/ +55%; Sonnet +283%/ +54%; Opus +229%/ +164%; GPT-5.5 +281%/ +274%; DeepSeek +93%/ +166%; Kimi +203%/ +249%, given as `api_post_ok/api_argorder`). One might read this as “verbose thrashing” causing the failure, but the relationship is correlational and uncontrolled: on the *still-passing* encoding cell the same superset arm ranges from −73% to +41% across the same models, with no consistent direction, which undercuts a clean “verbosity causes collapse” account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

One qualification governs this whole subsection. The retained completions (Appendix C.1) rule out the packaging faults the screens test but do not identify what replaces them, and no draw in the 480 replies of the two collapse cells declined the task. That screen was never run on the three

models whose draws were not retained, and elsewhere in the retained set it is not empty: it matches seven no-context Opus draws.

The claim structure is deliberately mechanism-agnostic: Theorem 3 verifies that the realized rule is not Blackwell-coherent *whatever* the internal cause, and the behavioral controls constrain the hypothesis space from outside, the collapse survives reversal of the block order (order control) and is not reproduced by an off-topic block of equal character length (padding control), and is largely curable by explicit convention routing (instruction ablation), a signature consistent with mis-binding rather than capacity exhaustion. That the one-line routing note recovers most of the collapse ($0/40 \rightarrow 33/40$ and $0/40 \rightarrow 39/40$) while labeled XML segmentation of the same content recovers none of it ($0/40$ on both cells) locates the operative variable in convention-to-task routing rather than formatting; this is an interpretation, not a finding, and every verified claim is independent of it. An activation-level account is out of reach here, since four of the six models are closed-weight and the phenomenon is cross-vendor; the direct route is to reproduce the collapse on an open-weight model with the same harness and instrument the contrasting arms, downstream of rather than prerequisite to the behavioral claim. Distractor interference in RAG has been localized to a sparse feature subspace and corrected there [47]; ours differs in cause, the added source is each true, and each needed by some task in the battery though not by every task rather than a distractor, and it is not a knowledge conflict in the sense of Xu et al. [48], since nothing in W_1 and W_2 contradicts.

D Reproducibility and protocol

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The complete artifact bundle (the measurement harness, the executable verifiers, and every per-model run log and result JSON cited below) is available at <https://github.com/abilous-ti/blackwell-llm-context>. The harness (`harness/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a stdlib regularized incomplete beta), is cross-OS, and implements the surrogate $\delta_{\mathcal{D}}$, the uniform incomparability verification (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference verification, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic Messages API, the Azure Responses API, and OpenAI-compatible chat completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

Protocol. The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type); return code 0 is PASS. The four verified cells are `api_post_ok`, `api_argorder` (`memo` arrives via a parameter, forcing the keyword-only fact), `enc_amount`, and `trap_store_wire`; the arms are $\{\text{none}, W_1, W_2, W_{1+}\}$, with W_{pad} (character-matched neutral filler $++W_1$) added for the padding control. All six models are queried over HTTP (one request, one turn, the prompt instructing tool abstention rather than a flag enforcing it; code extracted from the returned text), and PASS is measured over $n = 40$ i.i.d. completions per cell at $\eta = 0.10$.

Decoding, seeding, and retries. All runs use the provider’s default decoding configuration: the harness sets no explicit temperature, top- p , or seed (the launchers are invoked without sampling overrides), so the n draws per cell are independent samples from each provider’s default decoder, each a fresh single-turn call with a 200-second timeout. This describes every run reported in the

paper. Earlier versions of this record reached the Anthropic models, the four behavioural controls and the second source pair through the command line; each has since been re-measured over the API path, and every number reported here comes from those runs, whose logs carry the [HTTP-API] marker, the raw-failure audit of Section 7.4.5 included: it reads the completions retained by those API runs (`results/retain_api/`), not the record they replaced.

The following concerns the command-line record that the HTTP runs replaced; no table in this version is computed from those files, and the one re-measurement still in use is the GPT-5.5 trap cell described below. Four arm-only re-measurements were originally produced by ad-hoc scripts that were not kept, so their regime could be asserted but not recovered from a run log. All four were re-measured with `harness/diag/rerun_arm.py`, which calls the same `run_one` the six-model tables are built from and writes a log recording the regime. Three reproduced their published values closely: the Opus W_2 /trap_store_wire and Haiku W_1 /enc_amount cells returned 0/40 exactly as before, and the structured-context arm moved by one draw on one cell. The Opus W_{1+} arm moved further, and deeper: 38% $\rightarrow$ 12% on `api_post_ok` and 15% $\rightarrow$ 5% on `api_argorder`. Every run cited here has a log naming its regime; `harness/diag/check_regimes.py` lists them with the evidence for each—an inventory of the listed runs, not a proof that the list is complete or that each regime is disclosed where its numbers are used. The between-run movement on the Opus arm is itself a data point for Section 8.4: repeated measurement of the same cell months apart shifts by more than the nominal interval width. Transport-level failures (empty or zero-token results) are retried up to three times with quadratic backoff; a draw that still fails is scored as a failure, never as a pass. Where a transport-outage window hit a whole arm or a single cell, we re-measured it cleanly in isolation with the retry harness and report only the clean measurement: the Opus W_{1+} arm; the Opus W_2 /trap_store_wire cell, roughly 18 of whose 40 first-run draws were outage-scored (inferred from per-draw cost); and the Haiku W_1 /enc_amount cell, whose first run predated the retry wrapper. One cell resisted re-measurement for longer than the others: GPT-5.5 W_2 /trap_store_wire returned HTTP 500 after retries during the original run and no GPT-5.5 deployment was available on the resource at first revision, so for one revision cycle its reported 0/40 was carried with outage-scored draws in it. A GPT-5.5 deployment was subsequently provisioned and all four GPT-5.5 cells were re-run single-shot at $n = 40$ per arm with retention and zero transport failures. The re-run reproduces the published grid exactly in all eight W_1/W_2 entries, the disputed cell included (W_2 /trap_store_wire 0/40 clean), so every cell in the coverage family is now a valid binomial sample and the reported n is the number of real draws throughout.

One cell is sourced separately. The base GPT-5.5 run logged an HTTP 500 on W_2 /trap_store_wire, so *both* conditions of that task, W_1 and W_2 , are read from the retained re-measurement with zero transport errors (`audit/audit_gpt55_trap_store_wire_*`), which returns 40/40 and 0/40 from 80 recorded replies. The two conditions therefore share a measurement window; comparisons involving GPT-5.5’s other tasks remain subject to the between-window stationarity assumption. The reported percentages are unchanged; what changes is which run they are read from.

Artifacts. Per-model run logs and result JSONs are in the artifact repository (<https://github.com/abilous-ti/blackwell-llm-context>) under the pattern `blackwell_*.log,json`: the six single-turn $n = 40$ model runs, each a complete file and each read whole except for the W_1 and W_2 conditions of GPT-5.5’s trap_store_wire task described above (`blackwell_haiku_api_n40`, `blackwell_sonnet_api_n40`, `blackwell_opus_api_n40`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, `blackwell_kimi_n40`), and the follow-up controls. All four behavioural controls share one run, `blackwell_controls_api_n40` (640 calls), whose arms are the padding filler, the reversed concatenation, the routing note and the XML

segmentation measured against the same W_1 and W_{1+} baselines; `blackwell_pair2_api_n40` carries the second source pair and every $L_{\mathcal{D}}$ and Δ_T reported for it; `reranker_trap_api_n100` is the LLM listwise reranker probe, which records each judgment so the halfway checkpoint needs no second run (`harness/measure_reranker_trap.py`); and `dense_trap.json` (dense/cross-encoder probe, deterministic local scoring of open-weight checkpoints at zero API cost; `harness/measure_dense_trap.py`). The full per-cell record is in `docs/EXPERIMENT-BLACKWELL.md` and the theory hardening (formalization and proof sketches) in `docs/BLACKWELL.md`.

Compute and cost. The analysed record is 5,260 model calls: 3,840 across the six-model grid (640 per model), 640 for the behavioural controls, 480 for the second source pair and 300 reranker judgments. The repository holds a good deal more—superseded command-line runs, arm-only re-measurements and diagnostics—none of which is analysed here. We do not report a spend figure: the HTTP endpoints used here return no cost field, so any total would be an estimate from token counts rather than a measurement. The dense/cross-encoder probe ran locally on open weights at zero API cost.

References

- [1] David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.
- [2] Kawin Ethayarajh, Yejin Choi, and Swabha Swayamdipta. Understanding dataset difficulty with $\mathcal{V}$ -usable information. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *PMLR*, pages 5988–6008, 2022.
- [3] Hengran Zhang, Minghao Tang, Keping Bi, and Jiafeng Guo. Beyond relevance: Utility-centric retrieval in the LLM era. *arXiv preprint arXiv:2604.08920*, 2026. SIGIR 2026 tutorial.
- [4] Hengran Zhang, Keping Bi, Jiafeng Guo, Jiaming Zhang, Shuaiqiang Wang, Dawei Yin, and Xueqi Cheng. LLM-specific utility for retrieval-augmented generation. *arXiv preprint arXiv:2510.11358*, 2025. CIKM 2026.
- [5] Xin Luna Dong, Barna Saha, and Divesh Srivastava. Less is more: Selecting sources wisely for integration. *Proceedings of the VLDB Endowment*, 6(2):37–48, 2012. URL <http://www.vldb.org/pvldb/vol6/p37-dong.pdf>. VLDB 2013.
- [6] Lovisa Hagström, Ercong Nie, Ruben Halifa, Helmut Schmid, Richard Johansson, and Alexander Junge. Language model re-rankers are fooled by lexical similarities. In *Proceedings of the Eighth Fact Extraction and VERification Workshop (FEVER)*, pages 18–33, Vienna, Austria, 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.fever-1.2/>.
- [7] Xinyi Zhu, Haoyang Li, Yongqi Zhang, and Lei Chen. Skyline retrieval meets set-cover chunk merging: A cost-effective RAG-sketch for long-context LLM QA. *Proceedings of the ACM on Management of Data*, 4(3):1–27, 2026. doi: 10.1145/3802111.
- [8] Pia Alpila. Context selection as approximate sufficiency: Le Cam deficiency and Kolmogorov’s structure function against a fixed decoder, 2026. Preprint, 25 June 2026.
- [9] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.

- [10] Carlo Emilio Bonferroni. Teoria statistica delle classi e calcolo delle probabilità. *Pubblicazioni del R. Istituto Superiore di Scienze Economiche e Commerciali di Firenze*, 8:3–62, 1936.
- [11] A. Dvoretzky, J. Kiefer, and J. Wolfowitz. Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *The Annals of Mathematical Statistics*, 27(3):642–669, 1956. doi: 10.1214/aoms/1177728174.
- [12] Yue Yu, Wei Ping, Zihan Liu, Boxin Wang, Jiaxuan You, Chao Zhang, Mohammad Shoeybi, and Bryan Catanzaro. RankRAG: Unifying context ranking with retrieval-augmented generation in LLMs. *arXiv preprint arXiv:2407.02485*, 2024. NeurIPS 2024.
- [13] Xiaowei Yuan, Zhao Yang, Ziyang Huang, Yequan Wang, Siqi Fan, Yiming Ju, Jun Zhao, and Kang Liu. Exploiting contextual knowledge in LLMs through V-usable information based layer enhancement. *arXiv preprint arXiv:2504.15630*, 2025. ACL 2025.
- [14] Deepro Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, Yizhe Zhang, and Tom Rainforth. BED-LLM: Intelligent information gathering with LLMs and bayesian experimental design. *arXiv preprint arXiv:2508.21184*, 2025. ICLR 2026.
- [15] Kwan Ho Ryan Chan, Yuyan Ge, Edgar Dobriban, Hamed Hassani, and René Vidal. Conformal information pursuit for interactively guiding large language models. *arXiv preprint arXiv:2507.03279*, 2025.
- [16] Shan Xie, Man Luo, Chadly Daniel Stern, Mengnan Du, and Lu Cheng. DemoShapley: Valuation of demonstrations for in-context learning. *arXiv preprint arXiv:2410.07523*, 2024.
- [17] M. S. Vinay, Minh-Hao Van, and Xintao Wu. In-context learning demonstration selection via influence analysis. *arXiv preprint arXiv:2402.11750*, 2024.
- [18] Ikhtiyor Nematov, Tarik Kalai, Elizaveta Kuzmenko, Gabriele Fugagnoli, Dimitris Sacharidis, Katja Hose, and Tomer Sagi. Source attribution in retrieval-augmented generation. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2025)*, 2025. doi: 10.1007/978-3-032-19099-4_23.
- [19] Lu Dai, Yijie Xu, Jinhui Ye, Hao Liu, and Hui Xiong. SePer: Measure retrieval utility through the lens of semantic perplexity reduction. In *International Conference on Learning Representations (ICLR)*, 2025.
- [20] Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. Sufficient context: A new lens on retrieval augmented generation systems. *arXiv preprint arXiv:2411.06037*, 2024. ICLR 2025.
- [21] Debashish Chakraborty, Eugene Yang, Daniel Khashabi, Dawn Lawrie, and Kevin Duh. Principled context engineering for RAG: Statistical guarantees via conformal prediction. *arXiv preprint arXiv:2511.17908*, 2025.
- [22] Markus Frohmann, Mahdiyar Alavi, Elizabeth Lingg, and Navid Rekabsaz. Equal ranking quality, different decisions: Training order-consistent LLM scorers. *arXiv preprint arXiv:2608.26762*, 2026.

- [23] Peter Auer, Chao-Kai Chiang, Ronald Ortner, and Madalina Drugan. Pareto front identification from stochastic bandit feedback. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 51 of *PMLR*, pages 939–947, 2016.
- [24] Akriti Jain and Aparna Garimella. Modeling contextual passage utility for multihop question answering. In *Proceedings of IJCNLP-AACL 2025*, 2025.
- [25] Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. Is ChatGPT good at search? Investigating large language models as re-ranking agents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- [26] Yi Jiang, Sendong Zhao, Jianbo Li, Bairui Hu, Yanrui Du, Haochun Wang, and Bing Qin. OptiSet: Unified optimizing set selection and ranking for retrieval-augmented generation. *arXiv preprint arXiv:2601.05027*, 2026.
- [27] Yongqi Fan, Yuxiang Chu, Zhentao Xia, Xiaoyang Chen, Jie Liu, Haijin Liang, Jin Ma, Ben He, Yingfei Sun, Jie Zhai, Dezhi Ye, and Tong Ruan. Rank4Gen: RAG-preference-aligned document set selection and ranking. *arXiv preprint arXiv:2601.11273*, 2026.
- [28] Jaime Carbonell and Jade Goldstein. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336, 1998. doi: 10.1145/290941.291025.
- [29] Jiacheng Ye, Zhiyong Wu, Jiangtao Feng, Tao Yu, and Lingpeng Kong. Compositional exemplars for in-context learning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [30] Manu Ghulyani, Arunabh Singh, Karan Bharadwaj, Ankit Nath, and Suranjan Goswami. PACMS: Submodular context selection as a pluggable engine for LLM agents. *arXiv preprint arXiv:2606.20047*, 2026.
- [31] Hai Huang. Directed information γ -covering: An information-theoretic framework for context engineering. *arXiv preprint arXiv:2510.00079*, 2025.
- [32] Stephen E. Robertson. The probability ranking principle in IR. *Journal of Documentation*, 33(4):294–304, 1977. doi: 10.1108/eb026647.
- [33] Norbert Fuhr. A probability ranking principle for interactive information retrieval. *Information Retrieval*, 11(3):251–265, 2008. doi: 10.1007/s10791-008-9045-0.
- [34] Jun Wang and Jianhan Zhu. Portfolio theory of information retrieval. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 115–122, 2009. doi: 10.1145/1571941.1571963.
- [35] Guido Zuccon and Leif Azzopardi. Using the quantum probability ranking principle to rank interdependent documents. In *Advances in Information Retrieval (ECIR 2010)*, *LNCS 5993*, pages 357–369, 2010. doi: 10.1007/978-3-642-12275-0_32.
- [36] Will LeVine and Bijan Varjavand. Relevance isn’t all you need: Scaling RAG systems with inference-time compute via multi-criteria reranking. *arXiv preprint arXiv:2504.07104*, 2025. ICLR 2025.

- [37] Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation. *arXiv preprint arXiv:2605.03534*, 2026.
- [38] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [39] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.
- [40] Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. The power of noise: Redefining retrieval for RAG systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.
- [41] Bowen Jin, Jinsung Yoon, Jiawei Han, and Sercan O. Arik. Long-context LLMs meet RAG: Overcoming challenges for long inputs in RAG. *arXiv preprint arXiv:2410.05983*, 2024.
- [42] Shahar Levy, Nir Mazor, Lihi Shalmon, Michael Hassid, and Gabriel Stanovsky. More documents, same length: Isolating the challenge of multiple documents in RAG. *arXiv preprint arXiv:2503.04388*, 2025.
- [43] Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A. Huerta, and Hao Peng. Context length alone hurts LLM performance despite perfect retrieval. *arXiv preprint arXiv:2510.05381*, 2025. Findings of EMNLP 2025.
- [44] Kelly Hong, Anton Troynikov, and Jeff Huber. Context rot: How increasing input tokens impacts LLM performance. Technical report, Chroma, 2025. URL <https://www.trychroma.com/research/context-rot>.
- [45] Ming Liu. More is not always better: Cross-component interference in LLM agent scaffolding. *arXiv preprint arXiv:2605.05716*, 2026.
- [46] Alon Halevy and Jane Dwivedi-Yu. Learnings from data integration for augmented language models. *arXiv preprint arXiv:2304.04576*, 2023.
- [47] Christian Giannetti, Giovanni Trappolini, Nicola Tonellotto, Fabrizio Silvestri, and Pietro Liò. The mechanics of interference: Defusing distractors in RAG via sparse autoencoder interventions. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026. URL <https://aclanthology.org/2026.findings-acl.583/>.
- [48] Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- [49] Dingzirui Wang, Xuanliang Zhang, Keyan Xu, Qingfu Zhu, Wanxiang Che, and Yang Deng. When does context help? error dynamics of contextual information in large language models. *arXiv preprint arXiv:2602.08294*, 2026.

- [50] Guang Zhao, Byung-Jun Yoon, Gilchan Park, Shantenu Jha, Shinjae Yoo, and Xiaoning Qian. Pareto prompt optimization. In *International Conference on Learning Representations (ICLR)*, 2025.
- [51] Pia Alpila. Context selection has been a solved problem since 1951, 2026. Expository companion to the preprint above.
- [52] Pia Alpila. LM Recall: Code and data for “context selection as approximate sufficiency”, 2026. Software and data artifact.
- [53] Shuran Zheng, Xuan Qi, Rui Ray Chen, Yongchan Kwon, and James Zou. Proper dataset valuation by pointwise mutual information. *arXiv preprint arXiv:2405.18253*, 2024.
- [54] William F. Bradley. Enhancing LLM evaluations: The garbling trick. *arXiv preprint arXiv:2411.01533*, 2024.
- [55] Loïc Fosse, Frederic Bechet, Benoit Favre, Géraldine Damnati, Gwénolé Lecorvé, Maxime Darrin, Philippe Formont, and Pablo Piantanida. Statistical deficiency for task inclusion estimation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers*, 2025.
- [56] Deniz Akdemir. Le Cam distortion: A decision-theoretic framework for robust transfer learning. *arXiv preprint arXiv:2512.23617*, 2025.
- [57] Brendan van Rooyen and Robert C. Williamson. Le Cam meets LeCun: Deficiency and generic feature learning. *arXiv preprint arXiv:1402.4884*, 2014.
- [58] Napoleon Paxton. The theorems of dr. david blackwell and their contributions to artificial intelligence. *arXiv preprint arXiv:2604.06621*, 2026.
- [59] Zheng Zhang, Cuong C. Nguyen, Kevin Wells, and Gustavo Carneiro. Multi-agent decision making: A Blackwell’s informativeness approach. *arXiv preprint arXiv:2605.06028*, 2026.
- [60] Omar Khattab and Matei Zaharia. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 39–48, 2020. doi: 10.1145/3397271.3401075.
- [61] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. REALM: Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [62] Weijia Shi, Sewon Min, Michihiro Yasunaga, Minjoon Seo, Rich James, Mike Lewis, Luke Zettlemoyer, and Wen-tau Yih. REPLUG: Retrieval-augmented black-box language models. *arXiv preprint arXiv:2301.12652*, 2023. NAACL 2024.
- [63] Erik N. Torgersen. *Comparison of Statistical Experiments*, volume 36 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1991. ISBN 9780521250306. doi: 10.1017/CBO9780511666353.
- [64] M. Ali Khan, Haomiao Yu, and Zhixiang Zhang. A simple proof of Blackwell’s theorem on the comparison of experiments for a general state space. *Economics Letters*, 247:112146, 2025. doi: 10.1016/j.econlet.2024.112146.

- [65] Nils Bertschinger and Johannes Rauh. The Blackwell relation defines no lattice. In *2014 IEEE International Symposium on Information Theory (ISIT)*, pages 2479–2483, 2014. doi: 10.1109/ISIT.2014.6875280.
- [66] Lovisa Hagström, Sara Vera Marjanović, Haeun Yu, Arnav Arora, Christina Lioma, Maria Maistro, Pepa Atanasova, and Isabelle Augenstein. A reality check on context utilisation for retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. Introduces the DRUID dataset.
- [67] Haotian Chen, Qingqing Long, Siyu Pu, Xiao Luo, Wei Ju, Meng Xiao, Yuanchun Zhou, Jianghua Zhao, and Xuezhi Wang. DeepEra: A deep evidence reranking agent for scientific retrieval-augmented generated question answering. *arXiv preprint arXiv:2601.16478*, 2026.
- [68] Lucien Le Cam. Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4):1419–1455, 1964. doi: 10.1214/aoms/1177700372.
- [69] Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer Series in Statistics. Springer, New York, 1986. doi: 10.1007/978-1-4612-4946-7.
- [70] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024.
- [71] Irena Gao, Percy Liang, and Carlos Guestrin. Model equality testing: Which model is this API serving? In *International Conference on Learning Representations (ICLR)*, 2025.
- [72] Sam Bowyer, Laurence Aitchison, and Desi R. Ivanova. Position: Don’t use the CLT in LLM evals with fewer than a few hundred datapoints. *arXiv preprint arXiv:2503.01747*, 2025. ICML 2025.